

MATH40003 - Linear Algebra and Groups

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Notes taken by

Autumn 2025 – Spring 2026

Compiled on June 16, 2026

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More details at <https://stones-y.github.io>.

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Introduction

Why do we study linear algebra?

- 2D and 3D Geometry
- Signal processing: audio and video samples, data and communications
- and many more!

Chapter 1

Fields

1.1 Fields

Definition 1.1 (Field). Let F be a set with two binary operations, addition $+: F \times F \rightarrow F$ and multiplication $\cdot: F \times F \rightarrow F$. We say that F is a *field* if it satisfies the following axioms for all $a, b, c \in F$:

Axioms for Addition $(F, +)$:

A1. Associativity: $(a + b) + c = a + (b + c)$

A2. Commutativity: $a + b = b + a$

A3. Additive Identity: There exists an element $0 \in F$ such that $a + 0 = a$.

A4. Additive Inverse: For each $a \in F$, there exists an element $-a \in F$ such that $a + (-a) = 0$.

Axioms for Multiplication $(F, \cdot)$:

M1. Associativity: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

M2. Commutativity: $a \cdot b = b \cdot a$

M3. Multiplicative Identity: There exists an element $1 \in F$, with $1 \neq 0$, such that $a \cdot 1 = a$.

M4. Multiplicative Inverse: For each $a \in F \setminus \{0\}$ there exists an element $a^{-1} \in F \setminus \{0\}$ such that $a \cdot a^{-1} = 1$.

Distributivity:

D1. $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

Remark. This is all the structure we need from the real numbers to do linear algebra. We can do linear algebra with many other fields, not just $\mathbb{R}$, like the finite field $\mathbb{F}_2$. Abstracting the scalars into an arbitrary field F allows for some rich and interesting results in linear algebra.

Example 1.2. The integers modulo p , $\mathbb{F}_p := \{0, 1, \dots, p-1\}$, where p is prime, form a field. Addition and multiplication are defined modulo p . All properties except the existence of a multiplicative inverse follow easily from the fact that $\mathbb{Z}$ is a commutative ring.

Lemma 1.3. In $\mathbb{F}_p$, $xy = 0 \iff x = 0$ or $y = 0$.

Proof. $xy = 0 \implies p \mid xy \implies p \mid x$ or $p \mid y$ (by Euclid's lemma). This means $x \equiv 0 \pmod{p}$ or $y \equiv 0 \pmod{p}$. $\square$

Theorem 1.4. For every $a \in \mathbb{F}_p \setminus \{0\}$ there exists $a^{-1} \in \mathbb{F}_p \setminus \{0\}$ such that $aa^{-1} = 1 = a^{-1}a$.

Proof. Given $a \in \mathbb{F}_p$, by commutativity we only need to find a $b \in \mathbb{F}_p$ such that $ab = 1$. Finding an explicit construction for b is difficult, so we instead show that the set of multiples $S := \{a \cdot 0, a \cdot 1, \dots, a \cdot (p-1)\}$ is exactly equal to $\mathbb{F}_p$.

It suffices to show that the p elements $a \cdot 0, a \cdot 1, \dots, a \cdot (p-1) \in \mathbb{F}_p$ are all distinct. Suppose $a \cdot c = a \cdot d$. Then $a \cdot (c - d) = 0$.

By our lemma, since $a \neq 0$ ($p \nmid a$), we must have $c - d = 0$, which implies $c = d$. Thus, multiplication by a maps distinct elements to distinct elements. Since there are p elements in $\mathbb{F}_p$, the set S covers all of $\mathbb{F}_p$. Because $1 \in \mathbb{F}_p$, there must be some element $b \in \{0, \dots, p-1\}$ such that $ab = 1$. $\square$

Corollary 1.5. $\mathbb{F}_p$ is a field.

Remark. The proof above relies on the Pigeonhole Principle. Because multiplication by a non-zero element a is injective on the finite set $\mathbb{F}_p$, it must also be surjective.

Definition 1.6 (Subfield). A subset $F \subseteq E$ of a field E is a *subfield* if $1 \in F$ and it is closed under arithmetic: whenever $a, b \in F$ we have $a + b, ab \in F$; if $a \in F$ then $-a \in F$; and if $a \in F \setminus \{0\}$ then $a^{-1} \in F$.

Remark. This definition is slightly redundant. To prove a subset $F \subseteq E$ is a subfield, it actually suffices to show closure under addition and multiplication, the existence of multiplicative inverses for non-zero elements, and that $-1 \in F$. Because if $-1 \in F$ and we have the closures above, then $(-1)(-1) = 1 \in F$ and $(-1) + 1 = 0 \in F$. Additive inverses follow from closure of addition with -1 .

Example 1.7. Some examples of fields and subfields are:

- $\mathbb{Q}[i] := \{a + bi : a, b \in \mathbb{Q}\} \subseteq \mathbb{C}$ is a field (the "Gaussian rationals"). It is a subfield of $\mathbb{C}$.
- $\mathbb{Q}[\sqrt{2}] := \{a + b\sqrt{2} : a, b \in \mathbb{Q}\} \subseteq \mathbb{R}$ is a field, and it is a subfield of $\mathbb{R}$ (exercise: show this).
- The field of rational functions $F(X) := \left\{ \frac{p(X)}{q(X)} : p, q \in F[X], q \neq 0 \right\}$ is a field. $\mathbb{Q}(x)$ is a subfield of $\mathbb{R}(x)$, for example.

1.2 Vector Spaces

Definition 1.8. The vector space F^n of n -tuples over a field F is defined as:

$$F^n := \left\{ \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} : x_1, \dots, x_n \in F \right\}$$

Remark. This is the primordial example of a vector space. Every finite-dimensional vector space is structurally identical (isomorphic) to F^n . The geometric intuition of $\mathbb{R}^n$ (arrows, scaling, and adding tip-to-tail) carries over algebraically to arbitrary fields.

Definition 1.9 (Commutative Group). A commutative group $(V, +)$ is a set equipped with a binary operation $+: V \times V \rightarrow V$ satisfying:

- **Associativity:** For all $u, v, w \in V$, $u + (v + w) = (u + v) + w$.
- **Commutativity:** For all $v, w \in V$, $v + w = w + v$.
- **Additive Identity:** There exists $0 := (0, \dots, 0) \in V$ such that for all $v \in V$, $v + 0 = 0 + v = v$.
- **Additive Inverse:** For all $v \in V$ there exists $-v \in V$ such that $v + (-v) = 0 = (-v) + v$.

Lemma 1.10. *In a commutative group $(V, +)$, the zero vector and additive inverses are unique.*

Proof. Given another zero vector $0'$, we have $0' = 0 + 0' = 0$. Thus, the zero vector is unique.

If w also acts as an additive inverse to v (so $w + v = 0$), then:

$$w = w + 0 = w + (v + (-v)) = (w + v) + (-v) = 0 + (-v) = -v.$$

Thus, the additive inverse is unique. $\square$

Definition 1.11 (Vector Space). A vector space V over a field F is a commutative group $(V, +)$ together with an action $F \times V \rightarrow V$, defined by $(\lambda, v) \mapsto \lambda v$, called *scalar multiplication*, satisfying the following properties for all $\lambda, \mu \in F$ and $v, w \in V$:

- **Associativity:** $\lambda(\mu v) = (\lambda\mu)v$
- **Distributivity of scalar sums:** $(\lambda + \mu)v = \lambda v + \mu v$
- **Distributivity of vector sums:** $\lambda(v + w) = \lambda v + \lambda w$
- **Unit multiplication:** $1 \cdot v = v$

For notation, unless stated otherwise, when we say “a vector space V ”, we assume by default that it is a vector space over a field F .

Remark. Structurally, one can think of a vector space as having two tiers: the “ground” field F provides the scalars (stretching factors), while the commutative group V provides the vectors (the objects being stretched and added; indeed, F acts on V to get scalar multiples of vectors). The axioms ensure these tiers interact in a linear way.

Example 1.12. Other examples of vector spaces include:

- $F[X] = \{a_0 + a_1X + \cdots + a_nX^n : a_i \in F\}$ (polynomials).
- Subfields $F \subseteq E$ (E acts as a vector space over its subfield F).
- $\text{Fun}(X, F) = \{f : X \rightarrow F\}$ (functions from an arbitrary set X into F).

Chapter 2

Matrices

2.1 Matrices and Linear Maps

Matrices are the “coordinated” way to describe linear transformations. We will formalise this in Theorem 2.3 below. But first, we define a linear transformation.

Definition 2.1 (Linear transformation). Fix a field F and two vector spaces V, W over F . A *linear transformation* (or linear map) is a function $T : V \rightarrow W$ such that:

- **Additivity:** $T(v_1 + v_2) = T(v_1) + T(v_2)$ for all $v_1, v_2 \in V$.
- **Homogeneity:** $T(\lambda v) = \lambda T(v)$ for all $\lambda \in F$ and $v \in V$.

Lemma 2.2. *If $T : V \rightarrow W$ is a linear transformation, then $T(0) = 0$.*

Proof. By linearity, $T(0) = T(0 + 0) = T(0) + T(0)$. Adding $-T(0)$ (the additive inverse of $T(0)$ in W) to both sides yields $T(0) = 0$. $\square$

Recall that any finite-dimensional vector space V , with dimension n , is isomorphic to F^n . For now, when we say we have written V and W in coordinates, we mean we replace V and W with F^n and F^m . This motivates the concept of matrices.

Theorem 2.3. *For all linear maps $T : F^n \rightarrow F^m$ defined by $(x_1, \dots, x_n) \mapsto (y_1, \dots, y_m)$, there exists an array of numbers (a_{ij}) such that:*

$$y_i = \sum_{j=1}^n a_{ij} x_j.$$

Proof. We define the standard elementary vectors $e_j \in F^n$ to have value 1 in the j -th entry, and 0 everywhere else. Since $T(e_j)$ is a vector in F^m , we can write it as a list of m coordinates. For all j , define $a_{1j}, \dots, a_{mj} \in F$ by:

$$T(e_j) = \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix}.$$

(Idea: think of $T(e_j)$ as the j^{th} column of the matrix (a_{ij})).

Any input vector can be written as $(x_1, \dots, x_n) = x_1 e_1 + \dots + x_n e_n$ by our definition of addition in F^n . By the linearity of T , we have:

$$\begin{aligned} T(x_1, \dots, x_n) &= T(x_1 e_1 + \dots + x_n e_n) \\ &= x_1 T(e_1) + \dots + x_n T(e_n) \\ &= \sum_{j=1}^n x_j \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix}. \end{aligned}$$

Taking the i^{th} row (coordinate) of the above sum, we get:

$$y_i = \sum_{j=1}^n a_{ij} x_j,$$

as required. □

Remark. The converse is also true: any function $T : F^n \rightarrow F^m$ given by such an array (a_{ij}) is a linear map. This is fundamental: *linear maps are exactly the functions that can be represented by matrices.*

Observe that all the information needed to compute this linear transformation is encoded in the array $A := (a_{ij})$.

Definition 2.4 (Matrix). The set of $m \times n$ matrices over a field F is defined as:

$$M_{m \times n}(F) := \left\{ \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} : a_{ij} \in F \right\}.$$

How do we conveniently encode the action of T using this? We define matrix-vector multiplication:

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$$

The upshot is that, given a linear map $T : F^n \rightarrow F^m$, we *define* matrix multiplication with a column vector precisely to execute this map.

2.2 Matrix Multiplication

2.2.1 Motivation

First, we motivate the definition of general matrix multiplication. Consider two linear maps:

- $T : F^k \rightarrow F^n$ represented by the matrix $B \in M_{n \times k}(F)$
- $S : F^n \rightarrow F^m$ represented by the matrix $A \in M_{m \times n}(F)$

Define the composite linear map $R : F^k \rightarrow F^m$ by $R = S \circ T$. By Claim ($\star$), this new linear map must also have a matrix representation, which we will call $C \in M_{m \times k}(F)$.

We *want* to calculate $c_{i\ell}$, the $(i, \ell)^{\text{th}}$ element of the matrix C . Let's calculate the ℓ^{th} column of C , call it C_ℓ . We know that each column of C represents the effect of the linear map R on the standard basis vector e_ℓ :

$$C_\ell = R(e_\ell) = S(T(e_\ell)).$$

We know $T(e_\ell)$ is exactly the ℓ^{th} column of B , call it B_ℓ . So we must calculate $S(B_\ell)$. Because B_ℓ is a column vector in F^n , evaluating $S(B_\ell)$ is equivalent to taking a linear combination of the columns of A (which we will denote $A_1, \dots, A_n$):

$$S(B_\ell) = b_{1\ell}A_1 + \cdots + b_{n\ell}A_n.$$

To find $c_{i\ell}$, we take the i^{th} row of this result:

$$\begin{aligned} c_{i\ell} &= (C_\ell)_i = b_{1\ell}(A_1)_i + \cdots + b_{n\ell}(A_n)_i \\ &= b_{1\ell}a_{i1} + \cdots + b_{n\ell}a_{in} \\ &= \sum_{j=1}^n a_{ij}b_{j\ell}. \end{aligned}$$

This is exactly the formula for $c_{i\ell}$ that we use to define matrix multiplication. Therefore, the fact that composition of linear maps gives the formula above for the numbers $c_{i\ell}$ is why we define matrix multiplication in the following way:

Definition 2.5 (Matrix operations).

- Matrix multiplication is the function $\cdot : M_{m \times n}(F) \times M_{n \times p}(F) \rightarrow M_{m \times p}(F)$ defined by:

$$(a_{ij})_{m \times n} (b_{kl})_{n \times p} := \left(c_{il} = \sum_{j=1}^n a_{ij} b_{jl} \right)_{m \times p}$$

- Matrix addition is the function $+ : M_{m \times n}(F) \times M_{m \times n}(F) \rightarrow M_{m \times n}(F)$ defined by:

$$(a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij}).$$

- Scalar multiplication for matrices is the function $\cdot : F \times M_{m \times n}(F) \rightarrow M_{m \times n}(F)$ defined by:

$$\lambda(a_{ij}) = (\lambda a_{ij}).$$

Remark. Just like $F^n \cong M_{n \times 1}(F)$, the set $M_{m \times n}(F)$ forms a vector space under matrix addition and scalar multiplication. However, matrix multiplication gives $M_{n \times n}(F)$ *additional* structure beyond just a vector space (it forms an algebra).

Definition 2.6 (Square Matrix). A matrix is square if $A \in M_{n \times n}(F)$ (the number of rows equals the number of columns). Square matrices of a fixed size can be multiplied by each other.

Proposition 2.7. *Matrix multiplication is associative: $(AB)C = A(BC)$.*

Proof.

$$((AB)C)_{il} = \sum_j (AB)_{ij} c_{jl} = \sum_j \left(\sum_k a_{ik} b_{kj} \right) c_{jl} = \sum_{j,k} a_{ik} b_{kj} c_{jl}$$

$$(A(BC))_{il} = \sum_k a_{ik} (BC)_{kl} = \sum_k a_{ik} \left(\sum_j b_{kj} c_{jl} \right) = \sum_{j,k} a_{ik} b_{kj} c_{jl}$$

Therefore, $((AB)C)_{il} = (A(BC))_{il}$. □

Corollary 2.8. *Given linear maps $S : F^m \rightarrow F^n$ and $T : F^n \rightarrow F^p$ with matrix representations A and B (so $S(v) = Av$ and $T(v) = Bv$), the matrix representation of the composite map $T \circ S : F^m \rightarrow F^p$ is given by the matrix product BA .*

Proof. $T \circ S(v) = T(S(v)) = B(Av) = (BA)v$ by associativity. $\square$

2.2.2 Some more formal properties of matrices

Definition 2.9 (Bilinear map). Let V, W, U be vector spaces over F . A map $f : V \times W \rightarrow U$ mapping $(v, w) \mapsto f(v, w)$ is *bilinear* if it is linear in each argument when the other is held fixed:

- $(\lambda_1 v_1 + \lambda_2 v_2, w) \mapsto \lambda_1 f(v_1, w) + \lambda_2 f(v_2, w)$
- $(v, \lambda_1 w_1 + \lambda_2 w_2) \mapsto \lambda_1 f(v, w_1) + \lambda_2 f(v, w_2)$

Remark. Note that this implies the standard scalar extraction property $f(\lambda v, w) = f(v, \lambda w) = \lambda f(v, w)$ (simply set $\lambda_2 = 0$).

Example 2.10. The dot product $F^n \times F^n \rightarrow F$, defined by $((a_1, \dots, a_n), (b_1, \dots, b_n)) \mapsto a_1 b_1 + \dots + a_n b_n$ is bilinear. This follows easily from the distributive properties of the field F .

We want to prove that matrix multiplication is bilinear. To do this, we first establish how matrix multiplication interacts with columns and rows.

Proposition 2.11. *Let $A \in M_{m \times n}(F)$ and $B = (B_1, \dots, B_p) \in M_{n \times p}(F)$, where B_j are the columns of B . Then:*

$$AB = (AB_1, \dots, AB_p)$$

Proof. By the formula, $(AB)_{i\ell} = \sum_k a_{ik} b_{k\ell}$. Notice that $\sum_k a_{ik} b_{k\ell}$ is exactly the i^{th} element of the matrix-vector product AB_ℓ . Thus, the ℓ^{th} column of AB is exactly the vector AB_ℓ . $\square$

Remark. This provides a *column-wise* way to calculate matrix products. To find the entire j^{th} column of AB , just calculate the matrix-vector product AB_j . The j^{th} column of AB **only** depends on the j^{th} column of B .

Proposition 2.12. *We can separate rows under right multiplication. Let B_i be the rows of B . Then:*

$$\begin{pmatrix} B_1 \\ \vdots \\ B_p \end{pmatrix} A = \begin{pmatrix} B_1 A \\ \vdots \\ B_p A \end{pmatrix}$$

Proof. Same logic as above, switching indices appropriately. $\square$

Corollary 2.13. *Matrix multiplication is bilinear.*

Proof. We prove linearity in the second argument. Let A be an $m \times n$ matrix, B and C be $n \times p$ matrices, and $\lambda \in F$.

$$\begin{aligned} A(\lambda B) &= A(\lambda B_1 \dots \lambda B_n) \\ &= (A(\lambda B_1) \dots A(\lambda B_n)) && \text{(Column separation)} \\ &= (\lambda(AB_1) \dots \lambda(AB_n)) && \text{(Matrix-vector linearity)} \\ &= \lambda(AB_1 \dots AB_n) \\ &= \lambda(AB) \end{aligned}$$

And for addition:

$$\begin{aligned} A(B + C) &= A(B_1 + C_1 \dots B_n + C_n) \\ &= (A(B_1 + C_1) \dots A(B_n + C_n)) \\ &= (A(B_1) + A(C_1) \dots A(B_n) + A(C_n)) \\ &= (A(B_1) \dots A(B_n)) + (A(C_1) \dots A(C_n)) \\ &= AB + AC \end{aligned}$$

Linearity in the first argument follows similarly. $\square$

Definition 2.14 (Transpose). The transpose of a matrix $A = (a_{ij}) \in M_{m \times n}(F)$ is defined as $A^\top := (a_{ji}) \in M_{n \times m}(F)$.

Proposition 2.15. *The transpose operation satisfies the following properties:*

- *Transpose is linear: $(A + B)^\top = A^\top + B^\top$ and $(\lambda A)^\top = \lambda A^\top$.*

- $(A^\top)^\top = A$.
- $(AB)^\top = B^\top A^\top$.

Proof. We prove the product rule. Let $C = AB$. Then $c_{ij} = \sum_k a_{ik}b_{kj}$. The (i, j) entry of $(AB)^\top$ is the (j, i) entry of AB , which is $\sum_k a_{jk}b_{ki}$. The (i, j) entry of $B^\top A^\top$ is the dot product of the i -th row of $B^\top$ and the j -th column of $A^\top$. This is exactly $\sum_k (B^\top)_{ik}(A^\top)_{kj} = \sum_k b_{ki}a_{jk} = \sum_k a_{jk}b_{ki}$. $\square$

Remark. Notice that the order reverses in $(AB)^\top = B^\top A^\top$. This is necessary for the dimensions to align. For example, if A is 1×5 and B is 5×1 , AB is 1×1 . $A^\top$ is 5×1 and $B^\top$ is 1×5 . The product $A^\top B^\top$ is impossible, but $B^\top A^\top$ results in a 1×1 matrix. There is a deeper reason for this, which we will see when we look at dual vector spaces and dual linear transformations.

2.2.3 Special Matrices

Definition 2.16 (Symmetric Matrix). A square matrix A is *symmetric* if $A = A^\top$.

Definition 2.17 (Orthogonal Matrix). A square matrix A is *orthogonal* if $AA^\top = I$.

Remark. For a square matrix, $AA^\top = I \iff A^\top A = I$. What does it mean for $AA^\top = I$? We will discuss this in more detail in a later chapter, but for now it is enough to note that if A_i are the rows of A , then $(AA^\top)_{ij} = A_i \cdot A_j$. Therefore:

$$A_i \cdot A_j = \delta_{ij} := \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

This means the *rows* of A are orthonormal. Similarly, $A^\top A = I$ implies the *columns* of A are orthonormal. Geometrically in $\mathbb{R}^n$, multiplying by an orthogonal matrix represents a rigid transformation (a rotation or reflection) that preserves lengths and angles (again, this will make more sense in a later chapter).

Definition 2.18 (Triangular and Diagonal Matrices). Let $A = (a_{ij})$ be a matrix.

- A is *upper-triangular* if $a_{ij} = 0$ for all $i > j$.

- A is *lower-triangular* if $a_{ij} = 0$ for all $i < j$.
- A is *diagonal* if it is both upper and lower triangular (i.e., $a_{ij} = 0$ for all $i \neq j$).

Chapter 3

Linear Equations

3.1 Solving Linear Systems

3.1.1 Gaussian Elimination

A system of m linear equations in n variables can be encoded by the matrix equation:

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}.$$

Definition 3.1 (Row equivalence). Let $A, B \in M_{m \times n}(F)$. We say that A and B are *row-equivalent* if B can be obtained from A by a finite sequence of the following elementary row operations:

- Permuting two rows.
- Multiplying a row by a non-zero scalar.
- Adding a scalar multiple of one row to another row.

Definition 3.2 (Row Echelon Form (REF)). A matrix is in *row echelon form* if:

- All rows consisting entirely of zeros are at the bottom.
- The leading non-zero entries (pivots) of each non-zero row are 1.
- The pivots of each non-zero row occurs strictly to the right of the pivot in the previous row.

Remark. Gaussian elimination is an algorithm that uses elementary row operations to transform any matrix into REF. This allows us to solve linear systems by back-substitution.

Example 3.3. Consider the augmented matrix of a linear system:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 5 & 3 & 10 \\ -1 & -1 & 1 & -1 \end{array} \right)$$

We apply Gaussian elimination to transform it into REF:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \end{array} \right) \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 + R_1 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right) R_3 \rightarrow R_3 - R_2$$

This matrix is now in REF.

3.1.2 Gauss-Jordan Elimination

Definition 3.4 (Reduced Row Echelon Form (RREF)). A matrix is in *reduced row echelon form* if:

- It is in row echelon form.
- Every pivot entry is equal to 1.
- Each pivot is the only non-zero entry in its column.

Remark. Gauss-Jordan elimination continues the process of Gaussian elimination to transform a matrix from REF into RREF. In RREF, the pivot columns are exactly the standard basis vectors e_i . Solving a system in RREF requires no back-substitution; each pivot equation directly expresses a pivot variable in terms of the free variables.

Example 3.5. Continuing from the previous example, we apply Gauss-Jordan elimination to reach RREF:

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right) \begin{array}{l} R_1 \rightarrow R_1 - R_3 \\ R_2 \rightarrow R_2 - R_3 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right) R_1 \rightarrow R_1 - 2R_2$$

The matrix is now in RREF. The solution is uniquely determined: $x_1 = 1, x_2 = 1, x_3 = 1$.

Theorem 3.6. *Every matrix is row-equivalent to a matrix in RREF.*

Proof. This is achieved constructively by the Gauss-Jordan elimination algorithm. $\square$

3.1.3 Uniqueness of RREF

Theorem 3.7. *An RREF augmented matrix $(A|b)$ represents an inconsistent system if and only if the last column is a pivot column.*

Proof. If the last column is a pivot column, the matrix contains a row of the form $(0 \cdots 0 \mid 1)$, which encodes the contradictory equation $0 = 1$. Thus, the system is inconsistent.

Conversely, if the system is inconsistent, its RREF must contain a contradiction of the form $(0 \cdots 0 \mid c)$ for some $c \neq 0$. Scaling this row by c^{-1} leaves a 1 in the final column, making it a pivot column. $\square$

Theorem 3.8 (Equivalence of Linear Systems).

1. *If $(A|b)$ is row-equivalent to $(A'|b')$, then the systems $Ax = b$ and $A'x = b'$ have the same solution set.*
2. *Conversely, if $(A|b)$ and $(A'|b')$ are consistent augmented matrices in $M_{m \times (n+1)}(F)$ with the same solution set, they are row-equivalent.*
3. *For any matrix A , there is a unique RREF matrix row-equivalent to A .*

Proof. 1. Elementary row operations are reversible and preserve the validity of equations. Permuting rows changes only the order; scaling a row by $\lambda \neq 0$ creates an equivalent equation; adding λR_i to R_j creates a new equation whose solutions must still satisfy the original system.

2. If $(A|b)$ and $(A'|b')$ are consistent and have the same solutions, their respective RREF forms must also have the same solutions by (1). The RREF of a consistent system is uniquely determined by its solution set,

because the parameterized solution explicitly identifies which variables are free and uniquely defines the pivot variables in terms of them. Therefore, their RREF matrices must be identical, meaning they are row-equivalent to each other.

3. Let B and B' be two RREF matrices row-equivalent to A . By (1), they have the same solution set. By (2), they must be identical since the solution set uniquely determines RREF, so $B = B'$. Therefore, there exists a unique RREF matrix that is row-equivalent to A . $\square$

Definition 3.9 (Rank). The *rank* of a matrix is the number of pivot columns in its REF or RREF.

Remark. The uniqueness of the RREF ensures that the rank of a matrix is a well-defined property, independent of the sequence of row operations used to compute it. The number of free variables is intrinsic to the solution set. For $A \in M_{m \times n}(F)$, we write $\text{rank}(A)$ for this number.

Definition 3.10 (Nullspace and Nullity). For a matrix $A \in M_{m \times n}(F)$, the *nullspace* of A is the set of all solutions to the homogeneous equation $Ax = 0$:

$$\text{Null}(A) := \{x \in F^n : Ax = 0\}$$

The *nullity* of A is the dimension of this nullspace.

Remark. We will rigorously define the concept of *dimension* later, but for now the *nullity* of a matrix A can be thought of as the number of free variables in the solution set of $Ax = 0$.

Theorem 3.11 (Rank-Nullity Theorem for Matrices). Let $A \in M_{m \times n}(F)$. Then:

$$\text{rank}(A) + \text{nullity}(A) = n$$

Proof. Let R be the unique reduced row echelon form (RREF) of A . Since elementary row operations preserve the solution set of homogeneous systems, $Ax = 0 \iff Rx = 0$. Thus, $\text{Null}(A) = \text{Null}(R)$.

By definition, the rank of A , denoted r , is the number of pivot columns in R . Because A has exactly n columns, the variables $x_1, \dots, x_n$ are partitioned uniquely into r pivot variables and $n - r$ free variables.

To completely describe the solution set $\text{Null}(R)$, we express each of the r pivot variables strictly in terms of the $n - r$ free variables. Any solution

$x \in \text{Null}(A)$ is entirely and uniquely determined by assigning arbitrary values from the field F to these $n - r$ free variables.

We construct a basis for $\text{Null}(A)$ by isolating the $n - r$ free variables: set one free variable to 1 and the remaining free variables to 0, and solve for the pivot variables. Doing this for each free variable generates exactly $n - r$ linearly independent vectors that span the entire solution space.

Therefore, the dimension of the nullspace (the nullity) is identically the number of free variables:

$$\text{nullity}(A) = n - r$$

Rearranging this equation yields $\text{rank}(A) + \text{nullity}(A) = n$. $\square$

Remark. This proof links the abstract concept of vector space dimensions to solving linear equations. The Rank-Nullity Theorem is essentially an accounting law: every column in a matrix either provides a pivot (adding 1 to the rank) or corresponds to a free variable (adding 1 to the nullity), so their sum must be n , the number of columns.

3.1.4 Structure of the Solution Space

When we solve a system $Ax = b$ using Gauss-Jordan elimination, the row operations depend only on A , not b . Consequently, if we find a solution to $Ax = b$, we are simultaneously exploring the structure of the homogeneous system $Ax = 0$.

Lemma 3.12. *Let x_p be a particular solution to $Ax = b$. Then any solution z to $Ax = b$ can be written as $z = x_p + v$, where v is a solution to the homogeneous system $Ax = 0$.*

Proof. Suppose $Az = b$ and $Ax_p = b$. By the linearity of matrix multiplication:

$$A(z - x_p) = Az - Ax_p = b - b = 0.$$

Thus, $v := z - x_p$ is a solution to the homogeneous system $Ax = 0$. $\square$

Remark. Geometrically, the solution set to $Ax = 0$ (the nullspace of A) forms a linear subspace (e.g., a line or plane through the origin). The solution set to $Ax = b$ is exactly this same subspace, translated away from the origin by the particular solution vector x_p . The homogeneous solutions define the “directions” of the solution space, and x_p acts as the “anchor point”.

3.2 Matrix Inverses

3.2.1 Matrix Equations

Now we consider inverting matrices. To solve multiple systems $Ax_1 = B_1, \dots, Ax_k = B_k$ simultaneously, we do not need to apply the row reduction algorithm k times. We can solve them all at once using an augmented matrix:

$$\left(A \mid B_1 \quad \cdots \quad B_k \right) \xrightarrow{\text{Gauss-Jordan}} \left(R \mid C_1 \quad \cdots \quad C_k \right)$$

where R is in RREF and $Rx_i = C_i$ for all i .

Remark. The row operations performed during Gauss-Jordan elimination depend only on the matrix A . The target columns $B_1, \dots, B_k$ are just transformed alongside A . Since matrix multiplication acts column-wise, $R(x_1 \dots x_k) = (Rx_1 \dots Rx_k)$, the concatenated problem is strictly equivalent to the single matrix equation $AX = B$, where X and B are matrices. The j^{th} column of X is the unique solution for the j^{th} column of B .

We apply this to find the inverse of a matrix. To find a right inverse X such that $AX = I_m$, we solve the matrix equation using the augmented matrix $\left(A \mid I_m \right)$.

Proposition 3.13. *Let $A \in M_{m \times n}(F)$. If $AX = I_m$ has a solution, then $m \leq n$.*

Proof. Suppose $m > n$. Then A is taller than it is wide. When we perform Gauss-Jordan elimination:

$$\left(A \mid I_m \right) \sim \left(R \mid C \right)$$

Since R has n columns and is in RREF, it can have at most n pivots. Because $m > n$, R must have at least one row of zeros at the bottom. By assumption, the system $AX = I_m$ is consistent, meaning $RX = C$ is also consistent. Therefore, a zero row in R implies the corresponding row in C must also be zero (to avoid an equation of the form $0 = c \neq 0$).

However, elementary row operations are invertible. If we apply the exact same sequence of operations directly to I_m , we get C . Since I_m is already the unique RREF matrix row-equivalent to itself, and C has a zero row, we cannot row-reduce C back to I_m , contradicting that C is row-equivalent to I_m . Thus, $m \leq n$. $\square$

Corollary 3.14. *Let $A \in M_{m \times n}(F)$. If A has a right inverse $X \in M_{n \times m}(F)$ and a left inverse $Y \in M_{n \times m}(F)$ such that $AX = I_m$ and $YA = I_n$, then A is a square matrix (i.e., $m = n$).*

Proof. Since A has a right inverse X , the equation $AX = I_m$ has a solution. By our previous proposition, this implies $m \leq n$.

Since A has a left inverse Y , we have $YA = I_n$. Taking the transpose of both sides yields $A^t Y^t = I_n$. Here, $A^t \in M_{n \times m}(F)$ and $Y^t \in M_{m \times n}(F)$. Applying the same proposition to this new equation, the number of rows of A^t must be less than or equal to its number of columns, which gives $n \leq m$.

Since $m \leq n$ and $n \leq m$, we conclude $m = n$. $\square$

Remark. The trick above relies on transposing the *left* inverse equation $YA = I_n$ to force the dimensionality bound (Proposition 3.13) in the opposite direction.

Lemma 3.15. *If A has a right inverse X and a left inverse Y , then $X = Y$.*

Proof. By associativity of matrix multiplication:

$$Y = YI_m = Y(AX) = (YA)X = I_n X = X$$

$\square$

Corollary 3.16. *If $A \in M_{m \times n}(F)$ has both a right inverse and a left inverse, then A is square, and all left and right inverses are equal to a unique two-sided inverse, denoted A^{-1} .*

Remark. To find a right inverse X such that $AX = I_m$, row reduce $(A \mid I_m)$ to $(R \mid C)$ and solve $RX = C$. To find a left inverse Y such that $YA = I_n$, we cannot directly use row operations. Instead, we transpose the equation to $A^t Y^t = I_n$. We then solve this as a right inverse problem for A^t to find Y^t , and transpose the final result to obtain Y . For a square matrix A , if a right inverse exists, it is the unique two-sided inverse A^{-1} , so we only need to perform Gauss-Jordan elimination once.

Theorem 3.17. *If $A \in M_{n \times n}(F)$ is invertible, then $(A \mid I_n)$ row reduces to $(I_n \mid A^{-1})$.*

Proof. Since A is invertible, $AX = I_n$ has a solution. Applying Gauss-Jordan elimination yields $(R \mid C)$, where $RX = C$ is consistent. Since R is square, if $R \neq I_n$, it must contain a zero row at the bottom. By consistency, the corresponding row in C would also be zero. But C is row-equivalent to I_n , which is impossible if C has a zero row. Therefore, $R = I_n$.

The system becomes $I_n X = C$, meaning $X = C$. Since the inverse is unique, $C = A^{-1}$. $\square$

Remark. We can alternatively prove Theorem 3.17 using elementary matrices. An elementary row operation on a matrix corresponds to left-multiplication by an elementary matrix E , where E is the matrix obtained from the identity by performing a single elementary row operation (exercise: prove that left multiplication by E corresponds to performing the same elementary row operation on A that was performed on E).

Now, let $A \in M_{n \times n}(F)$ be invertible, and consider the homogeneous system $Ax = 0$. Since A is invertible, we can left-multiply by A^{-1} to get $x = 0$. This means the system has only the trivial solution, which implies there are no free variables. An $n \times n$ matrix in RREF with no free variables must have a pivot in every column (by the Rank-Nullity Theorem for Matrices, Theorem 3.11), meaning its RREF is exactly I_n .

Because A row reduces to I_n , there exists a sequence of elementary row operations, corresponding to left-multiplication by elementary matrices $E_1, \dots, E_k$, such that:

$$E_k \cdots E_1 A = I_n$$

Let $E = E_k \cdots E_1$. Then $EA = I_n$. Multiplying both sides on the right by A^{-1} yields:

$$EAA^{-1} = I_nA^{-1} \implies E = A^{-1}$$

Applying this identical sequence of row operations to the augmented matrix $(A \mid I_n)$ corresponds to left-multiplying the entire block matrix by E :

$$E(A \mid I_n) = (EA \mid EI_n) = (I_n \mid E)$$

Since we established $E = A^{-1}$, the augmented matrix reduces exactly to $(I_n \mid A^{-1})$.

3.2.2 Preview of Determinants

When is a 2×2 matrix invertible? From the previous theorem, A is invertible if and only if its RREF is I_2 .

Proposition 3.18. *A 2×2 matrix A is invertible if and only if its rows R_1 and R_2 are not scalar multiples of each other.*

Proof. ($\Rightarrow$) If R_1 and R_2 are multiples of each other, row reduction will produce a row of zeros. Thus, the RREF is not I_2 , and A is not invertible.

($\Leftarrow$) If the rows are not multiples, then the first column cannot be entirely zero (otherwise R_1 and R_2 would be multiples of each other). After a possible row swap, we have:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \sim \begin{pmatrix} 1 & b' \\ 0 & d' \end{pmatrix}$$

where $a \neq 0$. Since R_2 is not a multiple of R_1 , the row reduction $R_2 \rightarrow R_2 - \frac{c}{a}R_1$ leaves a non-zero entry d' . Thus, the second column contains a pivot, making the RREF I_2 . $\square$

Remark. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the condition that the rows are not multiples means $ad \neq bc$. We define the determinant as $\det A := ad - bc$. Thus, A is invertible if and only if $\det A \neq 0$.

Chapter 4

Vector Spaces

4.1 Subspaces

Definition 4.1 (Subspace). Let V be a vector space over a field F . A subset $W \subseteq V$ is a *subspace* of V if it satisfies the following three conditions:

- $0 \in W$ (the additive identity is in W)
- Closure under scalar multiplication: $w \in W \implies \lambda w \in W$ for all $\lambda \in F$
- Closure under addition: $w_1, w_2 \in W \implies w_1 + w_2 \in W$

For notation, sometimes we write $W \leq V$ to indicate that $W \subseteq V$ is a subspace of V .

Remark. The condition $0 \in W$ ensures the subspace is not the empty set. Because vector spaces require closure under scalar multiplication, $0 \cdot w = 0$. Therefore, if W is non-empty (contains at least one element w), it must automatically contain 0 . Thus, the first condition can be replaced with $W \neq \emptyset$.

Definition 4.2 (Linear Combination). Let V be a vector space over a field F . A *linear combination* of vectors $v_1, \dots, v_k \in V$ is an expression of the form:

$$\lambda_1 v_1 + \dots + \lambda_k v_k \quad \text{where } \lambda_i \in F$$

Proposition 4.3. *Let V be a vector space over a field F , and let $v_1, \dots, v_k \in V$. Then*

$$W := \{\lambda_1 v_1 + \dots + \lambda_k v_k : \lambda_i \in F\} \subseteq V$$

is a subspace of V .

Proof. We use the definition of subspace. Setting all $\lambda_i = 0$ we get that $0 \in W$. Let $v, w \in W$. Then we have, for some $\lambda_i \in F, \mu_i \in F$,

$$v + w = \sum_{i=1}^k \lambda_i v_i + \sum_{i=1}^n \mu_i v_i = \sum_{i=1}^n (\lambda_i + \mu_i) v_i \in W.$$

Also, given $\mu \in F$ we have

$$\mu v = \mu \sum_{i=1}^n \lambda_i v_i = \sum_{i=1}^n (\mu \lambda_i) v_i \in W,$$

so W is a subspace of V . □

4.1.1 Matrix Subspaces

We can now define three subspaces associated with any matrix.

Definition 4.4 (Null Space, Row Space, Column Space). Let $A \in M_{m \times n}(F)$ with rows $R_1, \dots, R_m$ and columns $C_1, \dots, C_n$.

- **Null Space $N(A)$:** The solution set to the homogeneous equation $Ax = 0$.

$$N(A) := \{x \in F^n : Ax = 0\} \subseteq F^n$$

- **Row Space $RS(A)$:** The set of all linear combinations of the rows of A .

$$RS(A) := \{\lambda_1 R_1 + \dots + \lambda_m R_m : \lambda_i \in F\} \subseteq F^n$$

- **Column Space $CS(A)$:** The set of all linear combinations of the columns of A .

$$CS(A) := \{\lambda_1 C_1 + \dots + \lambda_n C_n : \lambda_i \in F\} \subseteq F^m$$

Proposition 4.5. $N(A)$ is a subspace of F^n .

Proof. Clearly $0 \in N(A)$ since $A0 = 0$. If $x \in N(A)$ and $\lambda \in F$, then $A(\lambda x) = \lambda(Ax) = \lambda 0 = 0$, so $\lambda x \in N(A)$. Finally, if $x, y \in N(A)$, then $A(x+y) = Ax + Ay = 0 + 0 = 0$, so $x+y \in N(A)$. Thus, $N(A)$ is a subspace of F^n . $\square$

Remark. Recall that, by linearity, the solutions to $Ax = b$ are exactly the solutions of $Ax = 0$ translated by a particular solution v (where $Av = b$). Thus, any two solutions differ only by a vector in the null space $N(A)$.

4.1.2 Operations on Subspaces

Proposition 4.6. Let V be a vector space, and let U, W be subspaces of V . Then:

1. $U + W := \{u + w : u \in U, w \in W\}$ is a subspace.
2. $U \cap W$ is a subspace.
3. $U \cup W$ is not a subspace unless $U \subseteq W$ or $W \subseteq U$.

Proof. 1. Take $u_1 + w_1$ and $u_2 + w_2$ in $U + W$. By commutativity and associativity of vector addition:

$$(u_1 + w_1) + (u_2 + w_2) = (u_1 + u_2) + (w_1 + w_2)$$

Since U and W are closed under addition, $u_1 + u_2 \in U$ and $w_1 + w_2 \in W$, meaning the sum remains in $U + W$. For scalar multiplication with $\lambda \in F$:

$$\lambda(u_1 + w_1) = \lambda u_1 + \lambda w_1$$

Since U and W are closed under scalar multiplication, $\lambda u_1 \in U$ and $\lambda w_1 \in W$, so the scaled vector is in $U + W$. Finally, $0 \in U$ and $0 \in W$, so $0 + 0 = 0 \in U + W$. Thus, $U + W$ is a subspace.

2. Take $v_1, v_2 \in U \cap W$ and $\lambda \in F$. Since $v_1, v_2 \in U$ and U is a subspace, $v_1 + v_2 \in U$ and $\lambda v_1 \in U$. By the exact same logic for W , $v_1 + v_2 \in W$ and $\lambda v_1 \in W$. Therefore, $v_1 + v_2$ and λv_1 are in both, so they are in $U \cap W$. Finally, $0 \in U$ and $0 \in W$, so $0 \in U \cap W$.

3. Exercise. $\square$

We can generalise these results to arbitrary collections of subspaces:

Proposition 4.7. Let $\{U_i\}_{i \in I}$ be a collection of subspaces of V .

- The intersection $\bigcap_{i \in I} U_i$ is a subspace.
- The sum $\sum_{i \in I} U_i := \{u_{i_1} + \cdots + u_{i_k} : k \in \mathbb{N}, i_j \in I, u_{i_j} \in U_{i_j}\}$ is a subspace.

Proof. The arguments for closure under addition, scalar multiplication, and the presence of the zero vector apply identically to the finite cases above. $\square$

Remark. In linear algebra, we only define finite sums of vectors to avoid issues of convergence. Therefore, the sum of an arbitrary family of subspaces $\sum_{i \in I} U_i$ must be defined specifically as the set of all possible *finite* sums of elements drawn from those subspaces.

4.2 Linear Independence and Span

Definition 4.8 (Linear Dependence). A list of vectors $(v_1, \dots, v_k)$ in V is *linearly dependent* if there exist scalars $\lambda_1, \dots, \lambda_k \in F$, not all zero, such that:

$$\lambda_1 v_1 + \cdots + \lambda_k v_k = 0$$

Otherwise, the list $(v_1, \dots, v_k)$ is *linearly independent*. Equivalently, they are independent if and only if the only solution to the above equation is the trivial solution $\lambda_1 = \cdots = \lambda_k = 0$.

Definition 4.9 (Span). The *span* of a list of vectors $(v_1, \dots, v_k)$ is the set of all their possible linear combinations:

$$\text{Span}(v_1, \dots, v_k) := \{\lambda_1 v_1 + \cdots + \lambda_k v_k : \lambda_i \in F\}$$

A list $(v_1, \dots, v_k)$ is called *spanning* for V if $\text{Span}(v_1, \dots, v_k) = V$.

Remark. By convention, the span of the empty list is the trivial subspace containing only the zero vector: $\text{Span}(\emptyset) := \{0\}$. This ensures that the span of any set is always a valid vector subspace.

Definition 4.10 (Finite-Dimensional). Let V be a vector space over a field F . We say that V is *finite-dimensional* if there exists a fi-

finite spanning set, i.e. there exists $v_1, \dots, v_k$, where $k \in \mathbb{N}$, such that $\text{Span}(v_1, \dots, v_k) = V$.

Remark. We define finite-dimensionality in terms of spanning sets rather than bases because the spanning condition is weaker. It allows us to establish the "size boundary" of the space without first needing to prove that a basis even exists.

Definition 4.11 (Basis). Let V be a vector space over a field F . A list of vectors $(v_1, \dots, v_k)$ is a *basis* for V if it is linearly independent and spans V .

Remark. Given these definitions, our goal now is to prove that every basis of a finite-dimensional space V has the exact same number of elements. If we have a spanning set where removing any v_i breaks the spanning property, that number n is unique. We will define this unique integer n as the dimension, denoted $\dim(V)$.

Remark. So far, we have defined span and linear independence for finite *tuples* of vectors $(v_1, \dots, v_n)$. We now redefine these concepts for *subsets* $X \subseteq V$.

Working with subsets allows X to be infinite without requiring infinite tuples. It also removes the need to track the ordering of elements.

Remark. Consider the tuple (v, v) and the set $\{v, v\} = \{v\}$ for $v \neq 0$. The tuple (v, v) is linearly dependent because $1v + (-1)v = 0$. However, the set $\{v\}$ is linearly independent.

When defining linear dependence for subsets, we must explicitly require that the vectors $x_1, \dots, x_n$ chosen from X are distinct. Otherwise, we could select the exact same element from the set multiple times to artificially create a non-trivial solution like $1v - 1v = 0$. For the definition of span, distinctness is not required because repeated vectors simply collapse into a single term: $\lambda_i v + \lambda_j v = (\lambda_i + \lambda_j)v$.

Definition 4.12. Let $X \subseteq V$. The *span* of X is the set of all finite linear combinations of elements of X :

$$\text{Span}(X) := \{\lambda_1 x_1 + \dots + \lambda_n x_n : n \in \mathbb{N}, x_i \in X, \lambda_i \in F\}$$

Lemma 4.13. *For any $X \subseteq V$, $\text{Span}(X)$ is a subspace of V .*

Proof. Follows from Proposition 4.3. □

Definition 4.14. A subset $X \subseteq V$ is *linearly dependent* if there exist distinct vectors $x_1, \dots, x_n \in X$ and scalars $\lambda_1, \dots, \lambda_n \in F$, not all zero, such that

$$\lambda_1 x_1 + \dots + \lambda_n x_n = 0.$$

Otherwise, X is *linearly independent*.

Definition 4.15. A subset $X \subseteq V$ is a *basis* for V if it is linearly independent and $\text{Span}(X) = V$.

Proposition 4.16. *Let V be a vector space, and let $v_1, \dots, v_n \in V$.*

- *The tuple $(v_1, \dots, v_n)$ is spanning $\iff$ the set $\{v_1, \dots, v_n\}$ is spanning.*
- *The tuple $(v_1, \dots, v_n)$ is linearly independent $\iff$ the vectors $v_1, \dots, v_n$ are distinct and the set $\{v_1, \dots, v_n\}$ is linearly independent.*

4.3 Dimension

Our goal want to prove that any two bases of the same finite-dimensional vector space have the exact same size. To do this, we will prove a stronger statement: given a finite-dimensional vector space V , for any linearly independent set $L \subseteq V$ and any finite spanning set $S \subseteq V$, we must have $|L| \leq |S|$. Because any two bases are both linearly independent and spanning, this will immediately force their sizes to be equal. First, we need to build some more theory:

4.3.1 Basic results on linear independence and span

Two simple observations that follow directly from the definitions are:

- If $X \subseteq Y$ and Y is linearly independent, then X is linearly independent.
- If $X \subseteq Y$ and X is spanning, then Y is spanning.

Now we establish how to grow or shrink sets without losing their key properties.

Lemma 4.17 (Extending linearly independent sets). *Suppose $X \subseteq V$ is linearly independent and $v \notin \text{Span}(X)$. Then $X \cup \{v\}$ is linearly independent.*

Proof. Suppose there exists a linear dependence:

$$\lambda v + \lambda_1 x_1 + \cdots + \lambda_n x_n = 0$$

where $x_1, \dots, x_n \in X$ are distinct, and the scalars $\lambda, \lambda_1, \dots, \lambda_n$ are not all zero.

- *Case 1:* $\lambda = 0$. Then $\lambda_1 x_1 + \cdots + \lambda_n x_n = 0$ with the remaining scalars not all zero. This is a linear dependence among vectors strictly in X , contradicting the linear independence of X .
- *Case 2:* $\lambda \neq 0$. We can solve for v :

$$v = -\frac{\lambda_1}{\lambda} x_1 - \cdots - \frac{\lambda_n}{\lambda} x_n$$

This implies $v \in \text{Span}(X)$, which contradicts our initial assumption.

Both cases yield a contradiction, so $X \cup \{v\}$ must be linearly independent. $\square$

Lemma 4.18 (Shrinking linearly dependent sets). *If $X \subseteq V$ is linearly dependent, then there exists $x \in X$ such that $\text{Span}(X) = \text{Span}(X \setminus \{x\})$.*

Proof. Because X is linearly dependent, there exists a non-trivial linear combination:

$$\lambda_1 x_1 + \cdots + \lambda_n x_n = 0$$

with at least one scalar $\lambda_i \neq 0$. Rearranging this equation to isolate x_i gives:

$$x_i = -\frac{\lambda_1}{\lambda_i} x_1 - \cdots - \frac{\lambda_n}{\lambda_i} x_n$$

where the right side is a linear combination of vectors in $X \setminus \{x_i\}$. Thus, $x_i \in \text{Span}(X \setminus \{x_i\})$, meaning its removal does not reduce the overall span of the set. $\square$

Combining these two ideas allows us to swap vectors in and out of a spanning set:

Lemma 4.19 (Steinitz Exchange Lemma). *Let $X \subseteq V$ and $v \in \text{Span}(X)$ with $v \neq 0$. Suppose:*

$$v = \lambda_1 x_1 + \cdots + \lambda_n x_n$$

for distinct $x_1, \dots, x_n \in X$ and all $\lambda_i \neq 0$ (we discard terms with zero coefficients). Then for any index i , we can exchange x_i for v without altering the span:

$$\text{Span}(X) = \text{Span}((X \setminus \{x_i\}) \cup \{v\})$$

Proof. Without loss of generality, assume $i = 1$. Because $\lambda_1 \neq 0$, we can rearrange the equation to isolate x_1 :

$$x_1 = \lambda_1^{-1}(v - \lambda_2 x_2 - \cdots - \lambda_n x_n)$$

This shows $x_1 \in \text{Span}((X \setminus \{x_1\}) \cup \{v\})$. Because x_1 is already in the span of the new modified set, adding it back in does not increase the span. Therefore, $\text{Span}((X \setminus \{x_1\}) \cup \{v\}) = \text{Span}(X \cup \{v\})$.

Since $v \in \text{Span}(X)$ by assumption, $\text{Span}(X \cup \{v\}) = \text{Span}(X)$. Combining these equalities yields the result. $\square$

4.3.2 Comparing Cardinalities

Lemma 4.20. *Let V be a finite-dimensional vector space. Let $X, Y \subseteq V$ be finite sets, with X linearly independent and Y spanning. There exists a subset $Y' \subseteq X \cup Y$ such that:*

$$Y' \supseteq X, \quad |Y'| = |Y|, \quad \text{and } Y' \text{ is spanning.}$$

Proof. We proceed by induction on $k := |X \setminus Y|$, the number of vectors in X that are not currently in Y .

Base Case: If $k = 0$, then $X \subseteq Y$. We simply set $Y' = Y$ and the conditions are trivially satisfied.

Inductive Step: Suppose $k > 0$ and the claim holds for $k - 1$. Choose a vector $x \in X \setminus Y$. Since Y is spanning, we can write x as a linear combination:

$$x = \lambda_1 y_1 + \cdots + \lambda_n y_n$$

with $y_i \in Y$ and all $\lambda_i \neq 0$.

We claim that at least one of these y_i is not in X . If every y_i were in X , we could rewrite the equation as:

$$-x + \lambda_1 y_1 + \cdots + \lambda_n y_n = 0$$

Since $x \notin Y$, x is distinct from all y_i . This creates a non-trivial linear combination of distinct elements of X equating to zero, which contradicts the linear independence of X . Thus, there exists some $y_i \notin X$.

Define a new set by swapping y_i out for x :

$$Y^* := (Y \setminus \{y_i\}) \cup \{x\}$$

By the Steinitz Exchange Lemma (Lemma 4.19), Y^* is still spanning. Because we removed an element not in X and added an element from X , the number of elements in X missing from our new spanning set has decreased by exactly one: $|X \setminus Y^*| = k - 1$. Furthermore, the size of the set remained constant: $|Y^*| = |Y|$.

By the inductive hypothesis applied to X and Y^* , there exists a spanning set Y' containing X such that $|Y'| = |Y^*| = |Y|$ and $Y' \subseteq X \cup Y^* \subseteq X \cup Y$. This completes the proof. $\square$

Remark. The induction here just formalises our algorithmic process of slowly putting each “problematic” $x \in X \setminus Y$ into a new set Y' by replacing a y_j which is outside of X (if it was already in X this is what we wanted - so replacing it would be redundant!). We then repeat and this process terminates because X is finite; this is what the induction formalises.

Consequently, we get:

Theorem 4.21. *For a vector space V , let $X \subseteq V$ be linearly independent and $Y \subseteq V$ be spanning. If $|Y| < \infty$, then $|X| \leq |Y|$.*

Proof. We first reduce to the case where $|X|$ is finite. Suppose $|Y| = m \in \mathbb{N}$. If $|X|$ were infinite, we could simply select a finite subset $X' \subseteq X$ such that $|X'| = m + 1$. Since X' is a subset of a linearly independent set, it is also linearly independent. If we can prove the theorem for finite sets, it would force $m + 1 \leq m$, a contradiction. Thus, we only need to prove the theorem assuming $|X| < \infty$.

But the theorem for $|X| < \infty$ follows from Lemma 4.20, which constructs a spanning set of size exactly $|Y|$ that completely contains X , so we are done. $\square$

4.3.3 Bases and Dimension

Theorem 4.22 (Invariance of Basis Size). *Any two bases of a finite-dimensional vector space V have the same cardinality.*

Proof. Let B_1 and B_2 be bases of V . Because V is finite-dimensional, there exists a finite spanning set $X \subseteq V$.

Since B_1 and B_2 are linearly independent and X is a finite spanning set, Theorem 4.21 guarantees that:

$$|B_1| \leq |X| \quad \text{and} \quad |B_2| \leq |X| < \infty$$

This proves both bases are finite. Now we apply the Theorem 4.21 again directly to the bases. Since B_1 is linearly independent and B_2 is spanning, $|B_1| \leq |B_2|$. Reversing their roles, since B_2 is linearly independent and B_1 is spanning, $|B_2| \leq |B_1|$. Therefore, $|B_1| = |B_2|$. $\square$

Before we can rigorously define dimension, we must prove that a basis actually exists in any finite-dimensional space. We can find a basis either by shrinking a spanning set or growing a linearly independent set, however since we *defined* finite-dimensional in terms of spanning sets, we need to prove the following fact:

Proposition 4.23. *Suppose $X \subseteq V$ is a minimal spanning set (meaning that for every proper subset $X' \subsetneq X$, X' does not span V). Then X is a basis of V .*

Proof. By assumption, X is spanning. We only need to show X is linearly independent.

Suppose for contradiction that X is linearly dependent. Then there exists a non-trivial linear combination:

$$\lambda_1 x_1 + \cdots + \lambda_n x_n = 0$$

with all $\lambda_i \neq 0$ among distinct elements $x_1, \dots, x_n \in X$.

By the Steinitz Exchange Lemma, we can exchange x_1 for the zero vector and maintain the exact same span. Thus, $(X \setminus \{x_1\}) \cup \{0\}$ spans V . Because the zero vector contributes nothing to a linear combination, the set $X \setminus \{x_1\}$ alone must span V .

This contradicts the minimality of X . Therefore, X must be linearly independent, making it a basis. $\square$

Corollary 4.24. *If $Y \subseteq V$ is a finite spanning set, then there exists a subset $X \subseteq Y$ which is a basis of V .*

Proof. Let Y be a spanning set with $|Y| < \infty$. If Y is linearly independent, it is already a basis and we are done.

If Y is linearly dependent, there exists a redundant vector $v \in Y$ such that $Y \setminus \{v\}$ still spans V . If this smaller set is still dependent, we repeat the process. Because Y is finite, this removal process must terminate. The resulting set $X \subseteq Y$ is a minimal spanning set, which is a basis by Proposition 4.23. $\square$

Remark. This proof is constructive. It provides an algorithm to extract a basis from a finite spanning set by systematically hunting for and removing linear dependencies. In practice, finding these dependencies is done by writing the vectors as columns in a matrix and applying Gaussian elimination to identify the pivot columns.

We have now proven two key facts for finite-dimensional vector spaces: at least one basis always exists, and all bases are exactly the same size. This allows us to define dimension rigorously:

Definition 4.25 (Dimension). If V is a finite-dimensional vector space, the *dimension* of V , denoted $\dim(V)$, is defined as the cardinality $|B|$ of any basis B of V .

Proposition 4.26. *If $X \subseteq V$ is a maximal linearly independent set (meaning no proper superset $Y \supsetneq X$ is linearly independent), then X is a basis of V .*

Proof. If X does not span V , there exists some $v \in V$ such that $v \notin \text{Span}(X)$. By our earlier extension lemma (Lemma 4.17), the set $X \cup \{v\}$ is linearly independent. This contradicts the maximality of X . Thus, every vector in V must lie in $\text{Span}(X)$, making X a spanning set and therefore a basis. $\square$

Corollary 4.27. *Let V be finite-dimensional. Let $X \subseteq V$ be linearly independent and $Y \subseteq V$ be spanning. There exists a basis B of V such that:*

$$X \subseteq B \subseteq X \cup Y$$

Proof. Let $n = \dim(V) < \infty$. If X spans V , it is already a basis, and we set $B = X$.

If X does not span V , then $\text{Span}(X) \subsetneq V = \text{Span}(Y)$. This implies Y cannot be entirely contained within $\text{Span}(X)$, so there exists some vector $y_1 \in Y$ such that $y_1 \notin \text{Span}(X)$.

By the extension lemma (Lemma 4.17), $X_1 := X \cup \{y_1\}$ is linearly independent. We repeat this process, obtaining a strictly increasing chain of linearly independent sets:

$$X = X_0 \subsetneq X_1 \subsetneq X_2 \subsetneq \cdots$$

where $X_k \subseteq X \cup Y$. Because V is finite-dimensional, the size of any linearly independent set is bounded by n . Thus, the process must terminate. The final set in this chain spans V , making it a basis B that contains X and is built entirely from elements of X and Y . $\square$

Remark. To formalize the “this process eventually terminates” argument, one would use induction on the integer $k := n - |X|$, representing the number of vectors needed to reach the dimension of the space.

The following proposition gives another useful way to think about bases.

Proposition 4.28. *Let V be a vector space and let $X, Y, B \subseteq V$.*

1. *X is linearly independent $\iff$ every vector $v \in V$ has at most one representation as a linear combination of elements of X .*
2. *Y is spanning $\iff$ every $v \in V$ has at least one representation as a linear combination of elements of Y .*
3. *B is a basis $\iff$ every $v \in V$ has exactly one representation as a linear combination of elements of B .*

Proof. 1. Suppose X is linearly independent and $v \in \text{Span}(X)$ has two expressions:

$$v = \lambda_1 x_1 + \cdots + \lambda_m x_m = \lambda'_1 x_1 + \cdots + \lambda'_m x_m,$$

for distinct $x_i \in X$. Subtracting the second from the first gives:

$$0 = (\lambda_1 - \lambda'_1)x_1 + \cdots + (\lambda_m - \lambda'_m)x_m.$$

Since X is linearly independent, the only solution to this homogeneous equation is the trivial one. This forces $\lambda_i - \lambda'_i = 0$, meaning $\lambda_i = \lambda'_i$ for all i . Thus, the representation is unique.

2. This is the exact definition of $\text{Span}(Y) = V$.
3. A basis is by definition both linearly independent and spanning. Combining (1) and (2) ensures both the existence and uniqueness of the linear combination. $\square$

Another useful proposition is:

Proposition 4.29. *Let V be a finite-dimensional vector space with $n := \dim(V)$. Then the following are equivalent for a subset $X \subseteq V$:*

1. X is a basis of V .
2. X is linearly independent and $|X| = n$.
3. X is spanning and $|X| = n$.

Proof. Clearly (1) implies both (2) and (3) by the definition of dimension. We are left to show that (2) implies (1) and (3) implies (1).

(2) $\Rightarrow$ (1): Let X be a linearly independent set with $|X| = n$. If X were not a basis, it would not span V . By our extension lemma, we could add a vector $v \notin \text{Span}(X)$ to obtain a strictly larger linearly independent set $X \cup \{v\}$ of size $n + 1$. However, no linearly independent set can be strictly larger than the dimension of the space, contradicting $\dim(V) = n$. Thus, X must span V .

(3) $\Rightarrow$ (1): Let X be a spanning set with $|X| = n$. If X were not linearly independent, we know from earlier lemmas there would exist a proper subset $X' \subsetneq X$ which is still spanning but with $|X'| < n$. The dimension of a space is the size of its minimal spanning sets (bases), so a spanning set of size strictly less than n contradicts $\dim(V) = n$. Thus, X must be linearly independent. $\square$

Remark. This is a powerful computational tool: if you know the dimension of a space is n , you only need to check *either* linear independence *or* spanning for a set of n vectors to prove it is a basis. You do not need to check both.

Example 4.30. $\dim(F^n) = n$ with the standard basis $\{e_1, \dots, e_n\}$.

Example 4.31. $\dim(M_{m \times n}(F)) = mn$. A natural basis is given by the matrices $E^{(i,j)}$ with a single 1 in position (i, j) and zeros elsewhere:

$$\left(E^{(i,j)}\right)_{k\ell} = \begin{cases} 1, & \text{if } k = i, \ell = j \\ 0, & \text{otherwise} \end{cases}$$

Structurally, $M_{m \times n}(F) \cong F^{mn}$; it is isomorphic to mn -tuples, we simply reorganize the entries into a rectangular array.

Example 4.32. In real geometry, $\dim(\mathbb{R}^3) = 3$. We can also look at dimensions of subspaces. A plane $U \subset \mathbb{R}^3$ passing through the origin has $\dim(U) = 2$ since $U = \text{Span}(u_1, u_2)$ where u_1, u_2 are linearly independent direction vectors. Similarly, a line $\ell \subset \mathbb{R}^3$ through the origin has $\dim(\ell) = 1$.

Note: Subspaces must pass through the origin to contain the zero vector. Lines or planes that do not pass through the origin are called *affine* subspaces (translations of vector subspaces).

4.3.4 Dimensions of Subspaces

Proposition 4.33. *If V is finite-dimensional and $U \subseteq V$ is a subspace, then U is also finite-dimensional and:*

$$\dim(U) \leq \dim(V).$$

Moreover, $\dim(U) = \dim(V)$ if and only if $U = V$.

Proof. Any basis of U is a linearly independent subset of V . We know the cardinality of any linearly independent set in V cannot exceed $\dim(V)$, giving $\dim(U) \leq \dim(V)$.

We must prove such a basis for U exists. Start with the empty set (which is linearly independent) and repeatedly adjoin vectors from U that are not yet in the span. Since no linearly independent subset of V can have more than $\dim(V)$ elements, this process must terminate in at most $\dim(V)$ steps, yielding a finite basis for U .

Finally, if $\dim(U) = \dim(V)$, then a basis of U is a linearly independent subset of V of size exactly $\dim(V)$. By our previous corollary, it must therefore be a basis for V , meaning its span is all of V , so $U = V$. Conversely, if $U = V$, clearly $\dim(U) = \dim(V)$. $\square$

Now, consider the following examples:

Example 4.34. Let $V = F^n$. Define the following subspaces:

$$U := \{(x_1, \dots, x_m, 0, \dots, 0) : x_i \in F\},$$

$$W := \{(0, \dots, 0, y_1, \dots, y_p) : y_i \in F\}.$$

Notice that U is isomorphic to F^m and W is isomorphic to F^p .

If $m + p \leq n$, there is no overlap between the non-zero entries. Thus, $U \cap W = \{0\}$, meaning $\dim(U \cap W) = 0$. The sum is:

$$U + W = \{(x_1, \dots, x_m, 0, \dots, 0, y_1, \dots, y_p) : x_i, y_j \in F\}.$$

Since $U + W \cong F^{m+p}$, we have $\dim(U + W) = m + p$. This matches the formula:

$$\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W) = m + p - 0 = m + p.$$

If $m + p \geq n$, the non-zero components overlap in exactly $m + p - n$ positions. Thus:

$$U \cap W \cong F^{m+p-n} \implies \dim(U \cap W) = m + p - n.$$

Because the combined non-zero components cover all n coordinates, $U + W = F^n$, so $\dim(U + W) = n$. This again matches the formula:

$$\dim(U) + \dim(W) - \dim(U \cap W) = m + p - (m + p - n) = n.$$

Example 4.35. Take two non-parallel planes $U, W \subset \mathbb{R}^3$ passing through the origin. Their sum $U + W$ is all of $\mathbb{R}^3$, so $\dim(U + W) = 3$. Their intersection $U \cap W$ is a 1-dimensional line. Notice that:

$$\dim(U + W) = 3 = 2 + 2 - 1 = \dim(U) + \dim(W) - \dim(U \cap W).$$

If we build a basis for $U + W$ by simply pasting together a basis for U and a basis for W , we will have 4 vectors, which is linearly dependent in $\mathbb{R}^3$. The redundancy exactly corresponds to the vectors that lie in the shared intersection $U \cap W$.

Given this motivation, we now state and prove the Dimension Formula for finite-dimensional vector spaces.

Theorem 4.36 (Dimension of Sums of Subspaces). *Let $U, W \subseteq V$ be finite-dimensional subspaces. Then:*

$$\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W).$$

Proof. Let $(v_1, \dots, v_m)$ be a basis for $U \cap W$. Because $U \cap W$ is a subspace of U , we can extend this to a basis of U by adding vectors $u_1, \dots, u_n$:

$$(v_1, \dots, v_m, u_1, \dots, u_n).$$

Similarly, we extend the intersection basis to a basis of W by adding vectors $w_1, \dots, w_p$:

$$(v_1, \dots, v_m, w_1, \dots, w_p).$$

We claim that the combined list:

$$B = (v_1, \dots, v_m, u_1, \dots, u_n, w_1, \dots, w_p)$$

is a basis for $U + W$.

Linear Independence: Suppose there is a linear dependence:

$$\sum_{i=1}^m \lambda_i v_i + \sum_{j=1}^n \mu_j u_j + \sum_{k=1}^p \nu_k w_k = 0$$

Rearranging to isolate the w_k terms:

$$\sum_{i=1}^m \lambda_i v_i + \sum_{j=1}^n \mu_j u_j = - \sum_{k=1}^p \nu_k w_k$$

Let this vector be denoted by x . The left side is a linear combination of basis vectors for U , so $x \in U$. The right side is a linear combination of vectors in W , so $x \in W$. Therefore, $x \in U \cap W$.

Because $x \in U \cap W$, it can be uniquely written in terms of the basis for the intersection:

$$x = \sum_{i=1}^m \alpha_i v_i$$

However, looking at our right side, x is written entirely using the w_k vectors:

$$\sum_{i=1}^m \alpha_i v_i = - \sum_{k=1}^p \nu_k w_k \implies \sum_{i=1}^m \alpha_i v_i + \sum_{k=1}^p \nu_k w_k = 0$$

Since $(v_1, \dots, v_m, w_1, \dots, w_p)$ is a basis for W , it is linearly independent. This forces all coefficients to be zero, meaning $\alpha_i = 0$ and $\nu_k = 0$.

Because $\nu_k = 0$, our original linear dependence collapses to:

$$\sum_{i=1}^m \lambda_i v_i + \sum_{j=1}^n \mu_j u_j = 0$$

Since $(v_1, \dots, v_m, u_1, \dots, u_n)$ is a basis for U , it is linearly independent. This forces $\lambda_i = 0$ and $\mu_j = 0$. Since all coefficients λ_i, μ_j, ν_k are exactly zero, the combined list B is linearly independent.

Spanning: Any vector in $U + W$ is of the form $u + w$. The vector u is in the span of the U basis, and w is in the span of the W basis. Adding them together simply creates a linear combination of vectors from the combined list B . Thus, B spans $U + W$.

We conclude that B is a basis for $U + W$. Counting the vectors, we have:

$$\dim(U + W) = m + n + p.$$

Observe that $\dim(U) = m + n$, $\dim(W) = m + p$, and $\dim(U \cap W) = m$, so we get

$$\begin{aligned} \dim(U) + \dim(W) - \dim(U \cap W) &= (m + n) + (m + p) - m \\ &= m + n + p \\ &= \dim(U + W), \end{aligned}$$

which concludes the proof. $\square$

Even though the union of vector spaces is not in general a subspace, we get the following natural result when we take the *span* of the union:

Lemma 4.37. *For subsets $X, Y \subseteq V$, $\text{Span}(X) + \text{Span}(Y) = \text{Span}(X \cup Y)$.*

Proof. The inclusion $\text{Span}(X) + \text{Span}(Y) \subseteq \text{Span}(X \cup Y)$ clearly holds because any sum of a vector in $\text{Span}(X)$ and a vector in $\text{Span}(Y)$ is just a linear combination of elements drawn from both sets, which lies in $X \cup Y$.

For the reverse inclusion, let $v \in \text{Span}(X \cup Y)$. It is a finite linear combination:

$$v = \lambda_1 z_1 + \cdots + \lambda_k z_k \quad \text{for } z_i \in X \cup Y.$$

We can group all terms where $z_i \in X$ and all terms where $z_i \in Y$ separately (arbitrarily assigning elements in the intersection to one side), to obtain:

$$v = (\mu_1 x_1 + \cdots + \mu_a x_a) + (\nu_1 y_1 + \cdots + \nu_b y_b),$$

where $x_i \in X$ and $y_j \in Y$. The first group lies entirely in $\text{Span}(X)$ and the second group lies entirely in $\text{Span}(Y)$. Thus, $v \in \text{Span}(X) + \text{Span}(Y)$. $\square$

4.3.5 Rank of a Matrix

We now revisit the row space, column space, null space, and rank of a matrix with our more sophisticated tools. For a matrix $A \in M_{m \times n}(F)$, recall that we denote these spaces as $\text{RS}(A)$, $\text{CS}(A)$, and $\text{Null}(A)$ respectively.

Proposition 4.38. *Let $A \in M_{m \times n}(F)$. Elementary row operations do not change the row space of A .*

Proof. If A' is obtained from A by a sequence of row operations, then each row of A' is a linear combination of the rows of A . Thus, $\text{RS}(A') \subseteq \text{RS}(A)$.

Conversely, since elementary row operations are invertible, each row of A can be reconstructed as a linear combination of the rows of A' . Hence, $\text{RS}(A) \subseteq \text{RS}(A')$. The two spaces coincide. $\square$

Corollary 4.39. *The dimension of the row space of $A \in M_{m \times n}(F)$ is equal to the number of non-zero rows in any row echelon form (REF) of A . This is identically the number of pivots.*

Proof. By the previous proposition, $\text{RS}(A) = \text{RS}(R)$, where R is the REF of A . The non-zero rows of R clearly span $\text{RS}(R)$. We only need to show they are linearly independent.

In REF, the leading non-zero entry (pivot) of each row occurs strictly to the right of the pivot in the row above it. Because of this "staircase" structure, no row can be constructed as a linear combination of the rows below it. Thus, any linear combination:

$$\lambda_1 R_1 + \cdots + \lambda_k R_k = 0$$

must evaluate to zero from top to bottom, forcing $\lambda_1 = 0$, then $\lambda_2 = 0$, and so on. The non-zero rows form a linearly independent spanning set for the row space. $\square$

Proposition 4.40. *Let $A \in M_{m \times n}(F)$ with columns $C_1, \dots, C_n$. The set of solutions to:*

$$\lambda_1 C_1 + \cdots + \lambda_n C_n = 0$$

is unaffected by row operations on A .

Proof. The linear combination can be written as the matrix-vector product $A\lambda = 0$, where $\lambda = (\lambda_1, \dots, \lambda_n)^\top$. If A' is obtained from A by row operations, we know from our study of linear systems that the homogeneous solution set is preserved:

$$A\lambda = 0 \iff A'\lambda = 0.$$

Thus, $\text{Null}(A) = \text{Null}(A')$. $\square$

Remark. This means elementary row operations do not change which subsets of columns are linearly independent. A subset of columns in A is linearly independent if and only if the corresponding subset of columns in A' is linearly independent.

Theorem 4.41 (Basis of the Column Space). *If the pivot columns of a row echelon form of A are at indices $i_1, \dots, i_k$, then the corresponding columns $\{C_{i_1}, \dots, C_{i_k}\}$ from the original matrix A form a basis for $\text{CS}(A)$.*

Proof. Let R be the reduced row echelon form (RREF) of A . In R , the pivot columns are exactly the standard basis vectors $e_1, \dots, e_k$. These columns are clearly linearly independent and they span the column space of R . Any non-pivot column in R is explicitly a linear combination of these pivot columns.

Because row operations preserve linear dependencies (as proven above), any linear relationship that exists among the columns of R exists identically among the columns of A .

Since the pivot columns of R are linearly independent, the corresponding columns of A are linearly independent. Since every non-pivot column of R is

a linear combination of the pivot columns of R , every non-pivot column of A is the exact same linear combination of the pivot columns of A .

Thus, the pivot columns of A are linearly independent and span $\text{CS}(A)$, making them a basis. $\square$

Example 4.42. Consider a matrix A and its RREF form R :

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 1 & 2 & 4 \end{pmatrix} \sim R = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Look at the column space of R . The pivot columns are $C'_1 = (1, 0, 0)^t$ and $C'_3 = (0, 1, 0)^t$. The second column is a redundant copy scaled by 2: $C'_2 = 2C'_1$.

Observation: Row operations completely change the column space. $\text{CS}(R)$ is the xy -plane, but $\text{CS}(A)$ is a different plane in $\mathbb{R}^3$. We cannot use C'_1 and C'_3 as a basis for $\text{CS}(A)$.

However, the *relationship* $C'_2 - 2C'_1 = 0$ translates to $R(-2, 1, 0)^t = 0$. Because $\text{Null}(R) = \text{Null}(A)$, it must be identically true that $A(-2, 1, 0)^t = 0$. Therefore, in the original matrix, $C_2 - 2C_1 = 0$, meaning $C_2 = 2C_1$.

Because the linear dependencies are identical, the positions of the independent columns (columns 1 and 3) remain the same. The basis for $\text{CS}(A)$ is the set of columns from the *original* matrix at those exact indices:

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \\ 4 \end{pmatrix} \right\}$$

Corollary 4.43. For any matrix $A \in M_{m \times n}(F)$, $\dim(\text{RS}(A)) = \dim(\text{CS}(A))$.

Proof. The dimension of the row space equals the number of non-zero rows in the RREF of A . Since every non-zero row contains exactly one pivot, this is equal to the number of pivot columns. By our previous theorem, the dimension of the column space is also exactly equal to the number of pivot columns. Thus, the dimensions are equal. $\square$

An alternative way to prove that for any $A \in M_{m \times n}(F)$, $\dim(\text{RS}(A)) = \dim(\text{CS}(A))$ (Corollary 4.43) is to use “matrix factorisations”; we provide this proof below:

Alternative proof. Let $k = \dim \text{CS}(A)$. Choose a basis $b_1, \dots, b_k$ for the column space of A . Form a matrix $B \in M_{m \times k}(F)$ whose columns are exactly these basis vectors.

Because every column A_j of A belongs to $\text{CS}(A)$, it can be written as a linear combination of the basis vectors:

$$A_j = c_{1j}b_1 + \cdots + c_{kj}b_k.$$

for some scalars $c_{ij} \in F$. We use these scalars to define a matrix $C \in M_{k \times n}(F)$ where the (i, j) -entry is c_{ij} .

By the column-wise definition of matrix multiplication, this explicitly means $A = BC$.

Now, consider the rows of A . By the row-wise view of matrix multiplication, the i^{th} row of A , denoted A_{i*} , is obtained by taking the dot product of the i^{th} row of B with the rows of C :

$$A_{i*} = b_{i1}C_{1*} + \cdots + b_{ik}C_{k*}$$

where C_{l*} denotes the l^{th} row of C .

This shows that every row of A is a linear combination of the k rows of C . Therefore, the row space of A is spanned by the rows of C . Since the dimension of a vector space is bounded by the size of any spanning set:

$$\dim \text{RS}(A) \leq k = \dim \text{CS}(A).$$

To obtain the reverse inequality, apply the exact same argument to the transpose $A^{\top}$. The column space of $A^{\top}$ is the row space of A , and the row space of $A^{\top}$ is the column space of A . Thus:

$$\dim \text{RS}(A^{\top}) \leq \dim \text{CS}(A^{\top}) \implies \dim \text{CS}(A) \leq \dim \text{RS}(A).$$

Combining both inequalities yields $\dim \text{RS}(A) = \dim \text{CS}(A)$. $\square$

Remark. This method is known as *rank factorization*. If a matrix A has column rank k , it can always be factored as $A = BC$, where the inner dimension is k . The idea is that the matrix factorization $A = BC$ reveals a dimensional “bottleneck”. When multiplying Ax , the operation can be split into two steps: $x \mapsto Cx \mapsto B(Cx)$. The matrix C compresses the n -dimensional input into a k -dimensional intermediate space, and B maps that k -dimensional intermediate space into the m -dimensional output space.

Because all information must pass through this k -dimensional bottleneck, the matrix cannot contain more than k independent linear equations (rows), nor can it reach more than k independent directions in the output (columns). The rank of a matrix is precisely the smallest inner integer k for which the factorization $A = BC$ exists.

Therefore, we can make the following definition:

Definition 4.44 (Rank). Let $A \in M_{m \times n}(F)$. The common dimension of the row space and the column space of A is called the *rank* of A and is denoted $\text{rank}(A)$.

The rank of $A \in M_{m \times n}(F)$ can thus be computed as the number of pivots. This is also the dimension of $\text{Span}(v_1, \dots, v_m)$ if we treat the vectors v_i either as rows or as columns. We now formalize the rank-nullity theorem using our new dimension terminology.

Theorem 4.45 (Rank-Nullity Theorem for Matrices). *For any matrix $A \in M_{m \times n}(F)$:*

$$\text{rank}(A) + \dim(\text{Null}(A)) = n.$$

Proof. Let k be the number of free variables in the homogeneous system $Ax = 0$. We know the solution set is parameterized by these k variables.

If R is the RREF of A and the free variables are $x_{i_1}, \dots, x_{i_k}$, we can construct specific solution vectors $v_1, \dots, v_k$ by setting exactly one free variable to 1 and the rest to 0. Every pivot variable is uniquely determined by these choices.

By construction, the list $(v_1, \dots, v_k)$ spans $\text{Null}(A)$. To see that it is linearly independent, consider the linear combination:

$$\lambda_1 v_1 + \dots + \lambda_k v_k = 0$$

Look specifically at the entries corresponding to the free variables x_{i_j} . Because each v_j has a 1 in the i_j^{th} position and a 0 in all other free variable positions, the i_j^{th} entry of the total sum is exactly λ_j . For the vector to be zero, we must have $\lambda_1 = \dots = \lambda_k = 0$.

Thus, $(v_1, \dots, v_k)$ is a basis for $\text{Null}(A)$, meaning $\dim(\text{Null}(A)) = k$. Since the number of pivot variables is exactly $\text{rank}(A)$, and the total number of variables is n :

$$\# \text{pivot variables} + \# \text{free variables} = n \implies \text{rank}(A) + \dim(\text{Null}(A)) = n,$$

as desired. $\square$

Theorem 4.46 (Invertibility Criteria). *The following are equivalent for a square matrix $A \in M_{n \times n}(F)$:*

1. A has a right inverse.
2. A has a left inverse.

3. $\text{rank}(A) = n$.

4. A is invertible.

Proof. We know from earlier that (3) is equivalent to (4) because $\text{rank}(A) = n \iff \text{RREF}(A) = I_n$. We also know that if A has both left and right inverses, it is invertible, so (4) implies (1) and (2). We only need to prove (1) $\Rightarrow$ (3) and (3) $\Rightarrow$ (1).

(1) $\Rightarrow$ (3): Suppose B is a right inverse of A , meaning $AB = I_n$. Notice that any column of AB is a linear combination of the columns of A . Therefore, the column space of the product is contained in the column space of the left matrix: $\text{CS}(AB) \subseteq \text{CS}(A)$. Because $AB = I_n$, we have $\text{CS}(AB) = \text{CS}(I_n) = F^n$. This forces $\text{CS}(A) = F^n$. Thus, $\text{rank}(A) = \dim(\text{CS}(A)) = n$.

(3) $\Rightarrow$ (1): Suppose $\text{rank}(A) = n$. Then $\text{CS}(A) = F^n$. Because the standard basis vectors e_i are in F^n , they are in the column space of A . This guarantees there exist vectors b_i such that $Ab_i = e_i$ for all $i = 1, \dots, n$. Sticking these column vectors together to form a matrix $B := (b_1 \dots b_n)$, we get:

$$AB = A(b_1 \dots b_n) = (Ab_1 \dots Ab_n) = (e_1 \dots e_n) = I_n$$

Thus, B is a right inverse.

By taking transposes, (2) $\iff$ (3) follows identically because $\text{rank}(A) = \text{rank}(A^\top)$. Thus, all statements are equivalent. $\square$

Remark. We can interpret the above result in terms of linear transformations. Recall that a matrix $A \in M_{m \times n}(F)$ encodes a linear transformation $T : F^n \rightarrow F^m$.

- Having a right inverse means T is surjective (onto), meaning its image (the column space) fills the entire codomain F^m .
- Having a left inverse means T is injective (one-to-one), meaning its kernel (the nullspace) is trivial $\{0\}$.

The Rank-Nullity Theorem states $\dim(\text{Image}) + \dim(\text{Kernel}) = n$. If the transformation is surjective, $\dim(\text{Image}) = m$, which forces $\dim(\text{Kernel}) = 0$ (injective). Conversely, if it is injective, $\dim(\text{Kernel}) = 0$, which forces $\dim(\text{Image}) = m$ (surjective). For finite-dimensional vector spaces of the same dimension, injectivity and surjectivity are strictly equivalent, just like in maps between finite sets of the same cardinality (we will see this later).

4.3.6 Computing Bases

Suppose we are given a subspace defined by a spanning set: $U = \text{Span}(v_1, \dots, v_k) \subseteq F^n$. Depending on the type of basis we want, we can extract one using Gauss-Jordan elimination in two different ways.

Method 1: Extracting a subset of the original vectors

- *Setup:* Form a matrix $A = (v_1 \dots v_k)$ where the vectors are the *columns*.
- *Action:* Row reduce A to its RREF, R .
- *Result:* Identify the pivot columns in R . The corresponding original columns of A form a basis for $\text{CS}(A) = U$.
- *Why?* Row operations preserve linear dependencies between columns. The pivot columns in R are a basis for $\text{CS}(R)$, so the corresponding columns in A are a basis for $\text{CS}(A)$.

Method 2: Constructing an echelon basis

- *Setup:* Form a matrix $A^\top$ where the vectors $v_1^t, \dots, v_k^t$ are the *rows*.
- *Action:* Row reduce $A^\top$ to its RREF, R' .
- *Result:* The non-zero rows of R' form a basis for $\text{RS}(A^\top) = U$.
- *Why?* Row operations do not change the row space ($\text{RS}(A^\top) = \text{RS}(R')$). The non-zero rows of an RREF matrix are guaranteed to be linearly independent due to their staircase structure. This creates a clean, simplified "echelon basis" that is highly useful for checking if a given vector lies in U .

Chapter 5

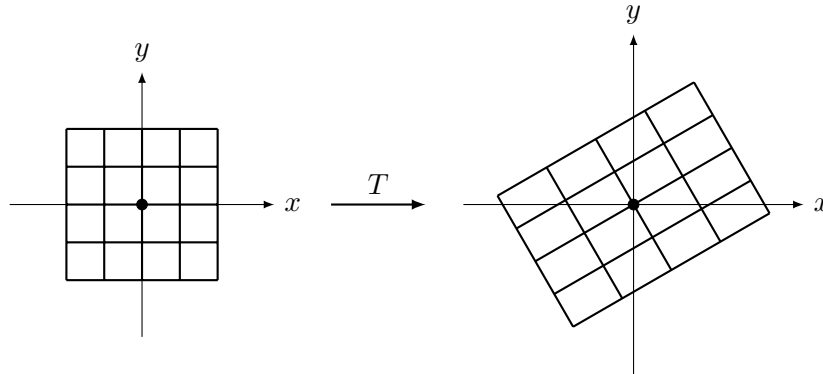
Linear Transformations

This is really what linear algebra is all about: linear maps between vector spaces.

Definition 5.1. Let $T : V \rightarrow W$ be a map between vector spaces over a field F . We say that T is *linear* if, for all $u, v \in V$ and $\lambda \in F$:

$$\begin{aligned}T(u + v) &= T(u) + T(v) \\T(\lambda v) &= \lambda T(v)\end{aligned}$$

This is an intuitive idea if one considers the case of linear maps between 2-dimensional Euclidean space, $\mathbb{R}^2$. The definition ensures that the 2D grid of squares generated by the standard basis (e_1, e_2) (i.e. the set of integer linear combinations of e_1 and e_2) is *preserved* in the sense that straight lines remain lines and the origin stays fixed. However, the generators e_1, e_2 may dilate or rotate, or become collinear depending on the linear map. These ideas are visualised in the diagram below. If one wants a more in-depth visualisation, 3Blue1Brown’s “Essence of Linear Algebra” playlist is excellent for acquiring geometric intuition (available at https://youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab&si=t4D-rnL5VLHZRyBi).



However, one must be careful to treat the geometric intuition of linear maps with caution, because in this course (and in mathematics in general) we would like to abstract and focus purely on structure; in this case, we abstract to linear maps between arbitrary vector spaces, in which *structure* rather than specific examples plays an important role in understanding. Nonetheless, particular examples, even in more exotic vector spaces, can help clarify concepts and bolster one's intuition.

An important result regarding how one can construct a linear map in practice is stated and proved below.

Proposition 5.2. *Let V and W be vector spaces over F , and let $(v_1, \dots, v_n)$ be a basis for V . For any arbitrary choice of vectors $w_1, \dots, w_n \in W$, there exists a unique linear map $T : V \rightarrow W$ such that:*

$$T(v_i) = w_i \quad \text{for all } i = 1, \dots, n$$

Proof. We must prove both that such a map exists (it is well-defined and linear) and that it is the only one.

Existence: Let $v \in V$. Because $(v_1, \dots, v_n)$ is a basis, v can be written uniquely as a linear combination:

$$v = \lambda_1 v_1 + \dots + \lambda_n v_n$$

We explicitly define the map T by taking those exact same scalars and applying them to the chosen vectors w_i :

$$T(v) := \lambda_1 w_1 + \dots + \lambda_n w_n$$

Because the representation of v is unique, this map is unambiguous (well-defined). To check that $T(v_i) = w_i$, notice that the unique representation of v_i has a coefficient of 1 on v_i and 0 on all other basis vectors. Plugging this into our definition immediately yields $1w_i = w_i$.

We now verify T is linear. Let $u = \alpha_1 v_1 + \cdots + \alpha_n v_n$.

$$\begin{aligned} T(v + u) &= T((\lambda_1 + \alpha_1)v_1 + \cdots + (\lambda_n + \alpha_n)v_n) \\ &= (\lambda_1 + \alpha_1)w_1 + \cdots + (\lambda_n + \alpha_n)w_n \\ &= (\lambda_1 w_1 + \cdots + \lambda_n w_n) + (\alpha_1 w_1 + \cdots + \alpha_n w_n) \\ &= T(v) + T(u) \end{aligned}$$

The proof for scalar multiplication $T(cv) = cT(v)$ follows identically:

$$T(cv) = T(c\lambda_1 v_1 + \cdots + c\lambda_n v_n) = c\lambda_1 w_1 + \cdots + c\lambda_n w_n = cT(v)$$

Thus, T is a valid linear map.

Uniqueness: Suppose there is another linear map $S : V \rightarrow W$ that also satisfies $S(v_i) = w_i$ for all i . Let $v = \lambda_1 v_1 + \cdots + \lambda_n v_n$ be an arbitrary vector in V . By the linearity of S :

$$\begin{aligned} S(v) &= S(\lambda_1 v_1 + \cdots + \lambda_n v_n) \\ &= \lambda_1 S(v_1) + \cdots + \lambda_n S(v_n) \\ &= \lambda_1 w_1 + \cdots + \lambda_n w_n \\ &= T(v) \end{aligned}$$

Since $S(v) = T(v)$ for all $v \in V$, the maps are identical ($S = T$). The map is uniquely determined by its action on the basis. $\square$

5.1 Isomorphisms

Definition 5.3. A linear map $T : V \rightarrow W$ is a (linear) *isomorphism* if it is bijective.

Proposition 5.4. Let V, W be vector spaces. If $T : V \rightarrow W$ is a linear isomorphism, its inverse $T^{-1} : W \rightarrow V$ is also a linear isomorphism.

Proof. Exercise. $\square$

Remark. Two vector spaces V, W being isomorphic (we write $V \cong W$) means they are structurally identical as vector spaces, even if their elements are completely different mathematical objects (such as polynomials, matrices, or column vectors). In the finite-dimensional case, an abstract vector space of dimension n can be viewed as a replica of F^n , as we will show below.

A linear isomorphism acts as a perfect translation dictionary between the two spaces. Because it is bijective, every element maps to exactly one

counterpart, meaning no information is lost or collapsed. Because it is linear, the algebraic structure is perfectly preserved: adding two vectors in V and then translating the result to W gives the exact same answer as translating them first and adding them in W . If $V \cong W$, any purely linear algebraic property (like dimension, linear independence, or span) true in V is automatically guaranteed to be true in W (we will show these facts later).

Theorem 5.5 (Coordinate Isomorphism). *Let V be a finite-dimensional vector space over a field F with $\dim(V) = n$. Then V is isomorphic to F^n .*

Proof. Because $\dim(V) = n$, there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ of size n .

By Proposition 4.28, every $v \in V$ has a unique representation $v = c_1v_1 + \dots + c_nv_n$. This allows us to construct a well-defined coordinate map $T : V \rightarrow F^n$ by extracting these unique scalars:

$$T(v) = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}$$

We claim that T is a linear bijection.

Linearity: Let $u, w \in V$ and $\lambda \in F$. Let their unique representations in $\mathcal{B}$ be $u = a_1v_1 + \dots + a_nv_n$ and $w = b_1v_1 + \dots + b_nv_n$. Adding the vectors gives:

$$u + w = (a_1 + b_1)v_1 + \dots + (a_n + b_n)v_n$$

Applying the map T to this sum:

$$T(u + w) = \begin{pmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{pmatrix} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} + \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = T(u) + T(w)$$

Similarly, for scalar multiplication:

$$\lambda u = (\lambda a_1)v_1 + \dots + (\lambda a_n)v_n$$

Applying T :

$$T(\lambda u) = \begin{pmatrix} \lambda a_1 \\ \vdots \\ \lambda a_n \end{pmatrix} = \lambda \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \lambda T(u)$$

Thus, T is a linear transformation.

Injectivity: Suppose $T(u) = T(w)$. This means their coordinate vectors in F^n are identical, meaning $a_i = b_i$ for all i . Therefore:

$$u = a_1v_1 + \dots + a_nv_n = b_1v_1 + \dots + b_nv_n = w$$

Since $T(u) = T(w) \implies u = w$, the map is injective.

Surjectivity: Let $x \in F^n$ be an arbitrary column vector with entries $x_1, \dots, x_n$. Define a vector $v \in V$ by:

$$v = x_1 v_1 + \dots + x_n v_n$$

By the definition of our map, $T(v) = x$. Since every vector in F^n has a pre-image in V , the map is surjective.

Since T is a linear transformation that is both injective and surjective, it is a linear isomorphism. Therefore, $V \cong F^n$. $\square$

Remark. For notation, the isomorphism $T : V \rightarrow F^n$ above is written $[\]_{\mathcal{B}} : V \rightarrow F^n$, where, for $v \in V$, $[v]_{\mathcal{B}}$ is the coordinate vector in F^n whose entries are the coefficients of v in the basis $\mathcal{B}$.

Example 5.6. Let $V = F[x]_{\leq 3}$ be the vector space of polynomials in F with degree at most 3, and let $\mathcal{B} = (1, x, x^2, x^3)$ be the standard basis of V . Then

$$[1 + 2x + 3x^2 + 4x^3]_{\mathcal{B}} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}.$$

From this, the comment regarding arbitrary finite-dimensional vector spaces V being replicas of $F^{\dim V}$ should be clear.

Proposition 5.7 (Isomorphisms preserve bases). *Let $T : V \rightarrow W$ be a linear isomorphism. If $(v_1, \dots, v_n)$ is a basis for V , then $(T(v_1), \dots, T(v_n))$ is a basis for W .*

Proof. We must show that the mapped list of vectors is both linearly independent and spans W .

Linear Independence: Suppose there is a linear combination equal to the zero vector in W :

$$\lambda_1 T(v_1) + \dots + \lambda_n T(v_n) = 0$$

By the linearity of T , we can combine the terms:

$$T(\lambda_1 v_1 + \dots + \lambda_n v_n) = 0$$

Because T is an isomorphism, it is injective. The only vector that maps to the zero vector under an injective linear map is the zero vector itself (we will prove this later). Thus:

$$\lambda_1 v_1 + \dots + \lambda_n v_n = 0$$

Since $(v_1, \dots, v_n)$ is a basis for V , it is linearly independent. This forces all the scalars to be zero: $\lambda_1 = \dots = \lambda_n = 0$. Therefore, $(T(v_1), \dots, T(v_n))$ is linearly independent.

Spanning: Let $w \in W$ be an arbitrary vector. Because T is an isomorphism, it is surjective, so there exists some $v \in V$ such that $T(v) = w$. Since $(v_1, \dots, v_n)$ spans V , we can express v as:

$$v = c_1 v_1 + \dots + c_n v_n$$

Applying T to both sides and using linearity yields:

$$w = T(c_1 v_1 + \dots + c_n v_n) = c_1 T(v_1) + \dots + c_n T(v_n)$$

This shows that every vector in W can be written as a linear combination of $(T(v_1), \dots, T(v_n))$. Thus, the list spans W .

Since the list is both linearly independent and spans the space, it is a basis for W . $\square$

As a corollary, we get:

Corollary 5.8. *Let V, W be finite-dimensional vector spaces. If $T : V \rightarrow W$ is a linear isomorphism then $\dim V = \dim W$.*

Remark. The structure of that proof reveals a useful dictionary between the properties of the linear map and the structural properties of the vector subsets. In general:

- A linear map is injective if and only if it sends linearly independent sets in V to linearly independent sets in W . It never “collapses” distinct independent directions into each other.
- A linear map is surjective if and only if it sends spanning sets of V to spanning sets of W . The mapped vectors have enough reach to cover the entire target space.

Since a basis is simply the combination of being linearly independent and spanning, and an isomorphism is the combination of being injective and surjective, it conceptually follows that isomorphisms must map bases exactly to bases.

5.2 Matrices and Linear Transformations

We have already explored the connection between matrices and linear maps in Chapter 2, however in this chapter we extend this theory to abstract

finite-dimensional vector spaces, not just F^n , using the idea that for any finite-dimensional vector space V , $V \cong F^{\dim V}$ from Theorem 5.5. We first look at the connection between linear maps and matrices for coordinate vector spaces F^n .

Proposition 5.9. *Let $T : F^n \rightarrow F^m$ be a map. Then T is linear if and only if there exists a unique matrix $A \in M_{m \times n}(F)$ such that $T(v) = Av$ for all $v \in F^n$ (treating elements of F^n and F^m as column vectors).*

Proof. Suppose T is linear. Let $(e_1, \dots, e_n)$ be the standard basis of F^n . Any $v \in F^n$ can be written uniquely as $v = \lambda_1 e_1 + \dots + \lambda_n e_n$. By linearity of T :

$$T(v) = T(\lambda_1 e_1 + \dots + \lambda_n e_n) = \lambda_1 T(e_1) + \dots + \lambda_n T(e_n)$$

Define $A \in M_{m \times n}(F)$ to be the matrix whose i^{th} column is $T(e_i)$, meaning $A := (T(e_1) \dots T(e_n))$. By the column-wise definition of matrix multiplication, $T(v) = Av$.

Conversely, if $A \in M_{m \times n}(F)$ and $T(v) = Av$, the linearity of T follows immediately from the linearity of matrix multiplication. The uniqueness of A follows because a linear map is entirely determined by its action on a basis. Since the columns $T(e_i)$ are fixed, the matrix A is uniquely determined. $\square$

Proposition 5.10. *Let V, W, U be vector spaces. If $T : V \rightarrow W$ and $S : W \rightarrow U$ are linear, then $S \circ T : V \rightarrow U$ is linear.*

Proof. Exercise. $\square$

5.2.1 Coordinate Vectors and Arbitrary Bases

We want to make precise how to write vectors and linear transformations in arbitrary bases. This process takes vectors in an abstract finite-dimensional vector space V to column vectors in F^n , and maps transformations between abstract spaces to standard matrices.

If $\mathcal{B} = (v_1, \dots, v_n)$ is a basis of V , there exists an isomorphism $F^n \rightarrow V$ given by:

$$(\lambda_1, \dots, \lambda_n) \mapsto \lambda_1 v_1 + \dots + \lambda_n v_n$$

This is a bijective linear map because every $v \in V$ has a unique representation in $\mathcal{B}$. The inverse isomorphism, denoted $[\]_{\mathcal{B}} : V \rightarrow F^n$, maps a vector to its coordinate column vector:

$$v \mapsto [v]_{\mathcal{B}} = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix}.$$

We call $[\cdot]_{\mathcal{B}}$ a *coordinate isomorphism*.

Definition 5.11 (Matrix of a Linear Map). Let $T : V \rightarrow W$ be a linear map between finite-dimensional vector spaces, with $\mathcal{B} = (v_1, \dots, v_n)$ a basis of V and $\mathcal{C} = (w_1, \dots, w_m)$ a basis of W . The *matrix of T with respect to $\mathcal{B}$ and $\mathcal{C}$* is the $m \times n$ matrix whose i^{th} column is the coordinate vector of $T(v_i)$ in the basis $\mathcal{C}$:

$${}_c[T]_{\mathcal{B}} := ([T(v_1)]_{\mathcal{C}} \dots [T(v_n)]_{\mathcal{C}})$$

Proposition 5.12. Let $T : V \rightarrow W$ be a linear map between finite-dimensional vector spaces. There exists a unique matrix ${}_c[T]_{\mathcal{B}} \in M_{m \times n}(F)$ such that for every $v \in V$:

$$[T(v)]_{\mathcal{C}} = {}_c[T]_{\mathcal{B}}[v]_{\mathcal{B}}$$

Equivalently, if $T(v_i) = \sum_{j=1}^m a_{ji}w_j$, then ${}_c[T]_{\mathcal{B}} = (a_{ji})$.

Proof. We first present a direct proof. We constructed ${}_c[T]_{\mathcal{B}}$ such that its i^{th} column is $[T(v_i)]_{\mathcal{C}}$. Thus:

$${}_c[T]_{\mathcal{B}}[v_i]_{\mathcal{B}} = {}_c[T]_{\mathcal{B}}e_i = [T(v_i)]_{\mathcal{C}}$$

The proposition holds exactly for the basis vectors v_i . Because any $v \in V$ is a unique linear combination of the basis vectors, and both coordinate mapping and matrix multiplication are linear operations, the equality holds for all $v \in V$. $\square$

Alternative proof. We now present a conceptual proof via coordinate isomorphisms. For $v \in V$, we have a unique coordinate vector $[v]_{\mathcal{B}} \in F^n$. Likewise, $T(v) \in W$ has a unique coordinate vector $[T(v)]_{\mathcal{C}} \in F^m$.

Define the composition map $\Phi : F^n \rightarrow F^m$ by:

$$\Phi([v]_{\mathcal{B}}) := [T(v)]_{\mathcal{C}} \implies \Phi = [\cdot]_{\mathcal{C}} \circ T \circ [\cdot]_{\mathcal{B}}^{-1}$$

Because $[\cdot]_{\mathcal{C}}$, T , and $[\cdot]_{\mathcal{B}}^{-1}$ are all linear maps, their composition Φ is a linear map from $F^n \rightarrow F^m$ by Proposition 5.10.

By our earlier proposition on coordinate spaces, every linear map from $F^n \rightarrow F^m$ is given by left-multiplication by a unique matrix. Let A be this matrix, so $\Phi = T_A$. Therefore:

$$[T(v)]_{\mathcal{C}} = \Phi([v]_{\mathcal{B}}) = A[v]_{\mathcal{B}}$$

To compute A , we evaluate Φ on the standard basis e_i :

$$Ae_i = \Phi(e_i) = [\cdot]_{\mathcal{C}} \circ T(v_i) = [T(v_i)]_{\mathcal{C}}$$

Thus, A matches our explicit construction of ${}_c[T]_{\mathcal{B}}$. $\square$

Note that the direct constructive proof showed existence, but the conceptual proof via coordinate isomorphisms showed *why* such a matrix *must* exist.

Remark. The conceptual proof is summarized by this commuting square diagram:

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ [\cdot]_{\mathcal{B}} \downarrow & & \downarrow [\cdot]_{\mathcal{C}} \\ F^n & \xrightarrow{T_A := \Phi} & F^m \end{array}$$

To compute the coordinates of $T(v)$ in W , we can travel across the top and down (apply T , then find coordinates), or travel down and across the bottom (find coordinates in V , then multiply by the matrix A). The diagram guarantees both paths yield the same result.

Example 5.13. Let $V = F[x]_{\leq m}$ and $W = F[x]_{\leq m-1}$ be the vector spaces of polynomials of degree at most m and $m-1$ respectively. Consider the derivative map:

$$T := \frac{d}{dx} : V \rightarrow W$$

With respect to the standard monomial bases $\mathcal{B} = (1, x, \dots, x^m)$ of V and $\mathcal{C} = (1, x, \dots, x^{m-1})$ of W , we map the basis vectors:

$$\frac{d}{dx}(x^k) = kx^{k-1}$$

The coordinate vector of kx^{k-1} in $\mathcal{C}$ has a k in the k^{th} position (since the index starts at x^0) and zeros elsewhere. Thus, the matrix is:

$${}_c[T]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & m \end{pmatrix}$$

Now consider the integration map $S : W \rightarrow V$:

$$\int x^k dx = \frac{x^{k+1}}{k+1} + c$$

With respect to the bases $\mathcal{C}$ and $\mathcal{B}$, the mapping of basis vectors is $x^k \mapsto c_k + \frac{1}{k+1}x^{k+1}$. The matrix encoding this transformation requires placing the constants $c_0, \dots, c_{m-1}$ in the first row (the x^0 coefficients):

$${}_{\mathcal{B}}[S]_{\mathcal{C}} = \begin{pmatrix} c_0 & c_1 & \cdots & c_{m-1} \\ 1 & 0 & \cdots & 0 \\ 0 & \frac{1}{2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{m} \end{pmatrix}$$

By matrix multiplication, one can check that ${}_c[T]_{\mathcal{B}\mathcal{B}}[S]_{\mathcal{C}} = I_m$. This algebraic result confirms that differentiation followed by integration returns the original polynomial in W (noting that the constants c_i are annihilated by the top row of zeros in the derivative matrix).

We now consider the composition of linear maps and its connection to matrix multiplication. We will see that the composition of linear maps corresponds exactly to the multiplication of their coordinate matrices.

Let $U \xrightarrow{S} V \xrightarrow{T} W$ be linear maps. Let the spaces have the following bases:

$$\begin{aligned} \mathcal{B} &= (u_1, \dots, u_m) \text{ for } U \\ \mathcal{C} &= (v_1, \dots, v_n) \text{ for } V \\ \mathcal{D} &= (w_1, \dots, w_p) \text{ for } W \end{aligned}$$

Proposition 5.14. *With the notation above, the matrix of the composed map is the product of the matrices:*

$${}_{\mathcal{D}}[T \circ S]_{\mathcal{B}} = {}_{\mathcal{D}}[T]_{\mathcal{C}} \cdot {}_{\mathcal{C}}[S]_{\mathcal{B}}$$

Proof. We evaluate $T \circ S$ on an arbitrary vector $u \in U$. By definition of the matrix of a linear map, we know that applying S gives:

$$[S(u)]_{\mathcal{C}} = {}_{\mathcal{C}}[S]_{\mathcal{B}}[u]_{\mathcal{B}}$$

Applying T to the resulting vector $S(u) \in V$ gives:

$$[T(S(u))]_{\mathcal{D}} = {}_{\mathcal{D}}[T]_{\mathcal{C}}[S(u)]_{\mathcal{C}}$$

Substituting the first equation into the second yields:

$$[(T \circ S)(u)]_{\mathcal{D}} = {}_{\mathcal{D}}[T]_{\mathcal{C}} ({}_{\mathcal{C}}[S]_{\mathcal{B}}[u]_{\mathcal{B}}) = ({}_{\mathcal{D}}[T]_{\mathcal{C}} {}_{\mathcal{C}}[S]_{\mathcal{B}}) [u]_{\mathcal{B}}$$

However, by defining the composed map $T \circ S$ directly, we also have:

$$[(T \circ S)(u)]_{\mathcal{D}} = {}_{\mathcal{D}}[T \circ S]_{\mathcal{B}}[u]_{\mathcal{B}}$$

Because these two equations hold for all $u \in U$, we can choose $u = u_i$, meaning $[u_i]_{\mathcal{B}} = e_i$. Multiplying by e_i extracts the i^{th} column of a matrix. Since the columns are identical for all i , the matrices must be identical:

$${}_{\mathcal{D}}[T \circ S]_{\mathcal{B}} = {}_{\mathcal{D}}[T]_{\mathcal{C}} \cdot {}_{\mathcal{C}}[S]_{\mathcal{B}}$$

□

Alternative proof. Using coordinate isomorphisms, we can represent each linear map as a transformation between standard coordinate spaces. The matrix representation is exactly the map that makes the diagram commute.

$${}_{\mathcal{D}}[T \circ S]_{\mathcal{B}} = [\cdot]_{\mathcal{D}} \circ (T \circ S) \circ [\cdot]_{\mathcal{B}}^{-1}$$

By inserting the identity map $I_V = [\cdot]_{\mathcal{C}}^{-1} \circ [\cdot]_{\mathcal{C}}$ in the middle, we get:

$${}_{\mathcal{D}}[T \circ S]_{\mathcal{B}} = ([\cdot]_{\mathcal{D}} \circ T \circ [\cdot]_{\mathcal{C}}^{-1}) \circ ([\cdot]_{\mathcal{C}} \circ S \circ [\cdot]_{\mathcal{B}}^{-1})$$

The first bracket is precisely the definition of ${}_{\mathcal{D}}[T]_{\mathcal{C}}$, and the second is ${}_{\mathcal{C}}[S]_{\mathcal{B}}$. Because the composition of transformations on coordinate spaces is defined by matrix multiplication, the result follows immediately. □

As an aside, we notice that the above formula for composition of linear maps and products of matrices seems more general than the original formula which works with only one linear map and one matrix. In fact, it is, and we can derive the original formula using this one, i.e. our original linear map matrix connection is a special case of the composition relationship!

Proposition 5.15. *Using notation as above, the formula*

$${}_{\mathcal{D}}[T \circ S]_{\mathcal{B}} = {}_{\mathcal{D}}[T]_{\mathcal{C}} \cdot {}_{\mathcal{C}}[S]_{\mathcal{B}},$$

implies our original formula

$$[T(v)]_{\mathcal{C}} = {}_{\mathcal{C}}[T]_{\mathcal{B}}[v]_{\mathcal{B}}.$$

Proof. The idea is that we want ${}_{\mathcal{C}}[S]_{\mathcal{B}}$ to be a column vector in F^n , so by definition S must be a map from a 1-dimensional vector space to an n -dimensional vector space.

Define

$$\text{Hom}(F, V) := \{S : F \rightarrow V : S \text{ is linear}\}.$$

A useful observation is that there exists a linear isomorphism $\varphi : \text{Hom}(F, V) \rightarrow V$ with $S \mapsto S(1) \in V$. Intuitively, we can think of each function in $\text{Hom}(F, V)$ as corresponding to a single “line” in V , and the function evaluated at 1 is the representative of that line, v itself. Any other output is just a scaled version of that vector by linearity, i.e. lies in “the direction of v ”.

To show that this is a linear isomorphism, if we are given $v \in V$ we can define $f_v \in \text{Hom}(F, V)$ by $f_v(\lambda) = \lambda v$, so that in particular $f_v(1) = v$, so the map is surjective, and the map is injective because for any $\lambda \in F$, by linearity we have $S(\lambda) = \lambda S(1)$ and $S'(\lambda) = \lambda S'(1)$ so if $S(1) = S'(1)$ then the functions are identical for all $\lambda \in F$, i.e. $S = S'$, so the map is injective, and by construction is linear, so is a linear isomorphism.

In other words, we can identify a vector $v \in V$ by its unique function in $\text{Hom}(F, V)$ evaluated at 1.

So, back to the proof, we have F as a 1-dimensional vector space with standard basis $\mathcal{E} := \{1\} \subset F$, so that $[1]_{\mathcal{E}} = (1)$, and let $v \in V$ be arbitrary. Now, recall that V is isomorphic to $\text{Hom}(F, V)$, so let $S \in \text{Hom}(F, V)$ be such that $S(1) = v$. Then by definition we have $_{\mathcal{B}}[S]_{(1)} = ([S(1)]_{\mathcal{B}})$, but $S(1) = v$ by construction so now we have, by the hypothesis,

$$c[(T \circ S)(1)]_{(1)} = [T(v)]_c = c[T]_{\mathcal{B}\mathcal{B}}[S]_{(1)}(1) = c[T]_{\mathcal{B}}[v]_{\mathcal{B}},$$

as required. $\square$

5.2.2 Change of Basis

We now introduce the “change-of-basis formula”, which allows us to translate the coordinate representation of a vector (or a linear map) from one basis to another.

Let $\mathcal{B}, \mathcal{B}'$ be two bases of V . We define the *change-of-basis matrix* as the matrix of the identity map $I_V : V \rightarrow V$ evaluated with respect to these two different bases:

$$_{\mathcal{B}'}[I_V]_{\mathcal{B}}$$

By the evaluation formula, this matrix satisfies:

$$[v]_{\mathcal{B}'} = _{\mathcal{B}'}[I_V]_{\mathcal{B}}[v]_{\mathcal{B}}$$

Remark. To compute the change-of-basis matrix $_{\mathcal{B}'}[I_V]_{\mathcal{B}}$, you take each basis vector from the “old” basis $\mathcal{B}$, find its coordinates in terms of the “new” basis $\mathcal{B}'$, and make those coordinate vectors the columns of your matrix.

When you multiply a coordinate vector $[v]_{\mathcal{B}}$ by this matrix, you are executing a systematic substitution: replacing every instance of an old basis vector with its exact expansion in the new basis vectors, and collecting the terms.

In other words, we can compute the change-of-basis matrix from $\mathcal{B}'$ to $\mathcal{B}$, denoted ${}_{\mathcal{B}}[I]_{\mathcal{B}'}$, as follows: if $\mathcal{B} := (v_1, \dots, v_n)$ and $\mathcal{B}' = (v'_1, \dots, v'_n)$ and we write

$$v'_i = a_{1i}v_1 + \dots + a_{ni}v_n,$$

then we have

$$v = \lambda_1 v'_1 + \dots + \lambda_n v'_n, \text{ i.e. } [v]_{\mathcal{B}'} = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix},$$

so

$$[v]_{\mathcal{B}} = \lambda_1 \begin{pmatrix} a_{11} \\ \vdots \\ a_{n1} \end{pmatrix}_B + \dots + \lambda_n \begin{pmatrix} a_{1n} \\ \vdots \\ a_{nn} \end{pmatrix}_B,$$

so by definition of matrix multiplication we get that

$${}_{\mathcal{B}}[I]_{\mathcal{B}'} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \in M_{n,n}(F),$$

whose i -th column is $[v'_i]_{\mathcal{B}}$.

Theorem 5.16 (Change of Basis for Linear Maps). *Let $T : V \rightarrow W$ be linear, with bases $\mathcal{B}, \mathcal{B}'$ for V and $\mathcal{C}, \mathcal{C}'$ for W . Then:*

$${}_{\mathcal{C}'}[T]_{\mathcal{B}'} = {}_{\mathcal{C}'}[I_W]_{\mathcal{C}} \cdot {}_{\mathcal{C}}[T]_{\mathcal{B}} \cdot {}_{\mathcal{B}}[I_V]_{\mathcal{B}'}$$

Proof. This follows directly from the composition of linear maps, since $T = I_W \circ T \circ I_V$.

For an explicit algebraic proof, let $v \in V$. First, we compute $[T(v)]_{\mathcal{C}'}$ by translating the target space coordinates:

$$[T(v)]_{\mathcal{C}'} = {}_{\mathcal{C}'}[I_W]_{\mathcal{C}} [T(v)]_{\mathcal{C}} = {}_{\mathcal{C}'}[I_W]_{\mathcal{C}} {}_{\mathcal{C}}[T]_{\mathcal{B}} [v]_{\mathcal{B}}$$

Now, substitute the source space coordinates $[v]_{\mathcal{B}} = {}_{\mathcal{B}}[I_V]_{\mathcal{B}'} [v]_{\mathcal{B}'}$:

$$[T(v)]_{\mathcal{C}'} = ({}_{\mathcal{C}'}[I_W]_{\mathcal{C}} {}_{\mathcal{C}}[T]_{\mathcal{B}} {}_{\mathcal{B}}[I_V]_{\mathcal{B}'}) [v]_{\mathcal{B}'}$$

However, by definition, applying T directly between the primed bases yields:

$$[T(v)]_{\mathcal{C}'} = {}_{\mathcal{C}'}[T]_{\mathcal{B}'} [v]_{\mathcal{B}'}$$

Because this holds for all coordinate vectors $[v]_{\mathcal{B}'}$, the matrix on the right must perfectly equal the grouped product of matrices on the left. $\square$

Remark. This formula reads right to left. To apply T to a vector given in $\mathcal{B}'$ coordinates and get the answer in $\mathcal{C}'$ coordinates:

1. Multiply by ${}_{\mathcal{B}}[I_V]_{\mathcal{B}'}$ to translate the input into the “old” $\mathcal{B}$ coordinates.
2. Multiply by ${}_{\mathcal{C}}[T]_{\mathcal{B}}$ to apply the transformation and get the output in “old” $\mathcal{C}$ coordinates.
3. Multiply by ${}_{\mathcal{C}'}[I_W]_{\mathcal{C}}$ to translate the output into the “new” $\mathcal{C}'$ coordinates.

Why is this useful? Well, we will see later that it is a lot easier to determine the image of basis vectors under T in special bases of a vector space, from which we can easily get ${}_{\mathcal{C}}[T]_{\mathcal{B}}$, which then allows us to find the matrix in terms of, for example, the standard basis. In other words, some bases are better suited for a linear map than others. One example is diagonalisation of a matrix; we choose our bases $\mathcal{B}, \mathcal{C}$ (in these cases $\mathcal{B} = \mathcal{C}$ since we are dealing with linear maps $V \rightarrow V$) to be such that the image of the basis vectors are merely stretched by the transformation, so that the matrix representing the linear transformation is diagonal, which has very nice properties (note that, of course, we are not guaranteed the existence of such a basis, so not all matrices / linear transformations are diagonalisable). We will develop this theory in a later chapter.

Example 5.17. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear map that reflects vectors across the line $y = 2x$. We want to find the standard matrix of this transformation, ${}_{\mathcal{E}}[T]_{\mathcal{E}}$, where $\mathcal{E} = (e_1, e_2)$ is the standard basis.

Finding this matrix directly using trigonometry or projection formulas is tedious. Instead, we can choose a new basis $\mathcal{B} = (v_1, v_2)$ that is “better suited” for the geometry of the reflection (a direction parallel to the line, and the normal direction):

- Let v_1 point along the line: $v_1 = (1, 2)^{\top}$. Since it is on the mirror line, $T(v_1) = v_1$.
- Let v_2 be orthogonal to the line: $v_2 = (-2, 1)^{\top}$. Since it is perpendicular to the mirror, it gets perfectly flipped, so $T(v_2) = -v_2$.

Because the transformation is so simple on these specific vectors, writing the matrix of T with respect to $\mathcal{B}$ is trivial. The coordinate vectors of the outputs are $[T(v_1)]_{\mathcal{B}} = (1, 0)^{\top}$ and $[T(v_2)]_{\mathcal{B}} = (0, -1)^{\top}$, giving the diagonal matrix:

$${}_{\mathcal{B}}[T]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Now, we use the change-of-basis formula:

$${}_{\mathcal{E}}[T]_{\mathcal{E}} = {}_{\mathcal{E}}[I]_{\mathcal{B}} \cdot {}_{\mathcal{B}}[T]_{\mathcal{B}} \cdot {}_{\mathcal{B}}[I]_{\mathcal{E}}$$

The matrix $\varepsilon[I]_{\mathcal{B}}$ simply contains the vectors of $\mathcal{B}$ written in standard coordinates as its columns:

$$\varepsilon[I]_{\mathcal{B}} = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$$

The matrix ${}_{\mathcal{B}}[I]_{\varepsilon}$ is its inverse, which we compute using the 2×2 inverse formula (determinant is $1(1) - (-2)(2) = 5$):

$${}_{\mathcal{B}}[I]_{\varepsilon} = \frac{1}{5} \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}$$

Multiplying them all together yields the standard matrix:

$$\begin{aligned} \varepsilon[T]_{\varepsilon} &= \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \frac{1}{5} \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix} \\ &= \frac{1}{5} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix} \\ &= \frac{1}{5} \begin{pmatrix} -3 & 4 \\ 4 & 3 \end{pmatrix} \end{aligned}$$

Example 5.18. Consider the space of 2×2 matrices $V = M_{2 \times 2}(F)$ and the space of polynomials in F with degree at most 2, $W = F[x]_{\leq 2}$. Let $T : V \rightarrow W$ be defined by extracting the trace and off-diagonal elements:

$$T \begin{pmatrix} a & b \\ c & d \end{pmatrix} = (a + d) + (b + c)x + (a - d)x^2$$

Let's find the matrix representation using the standard bases:

$$\mathcal{B} = \left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right) := (E_{11}, E_{12}, E_{21}, E_{22})$$

$$\mathcal{C} = (1, x, x^2)$$

Applying the coordinate isomorphism $[\cdot]_{\mathcal{C}}$, we evaluate T on each basis matrix in $\mathcal{B}$:

$$\begin{aligned} [T(E_{11})]_{\mathcal{C}} &= [1 + x^2]_{\mathcal{C}} = (1, 0, 1)^{\top}, \\ [T(E_{12})]_{\mathcal{C}} &= [x]_{\mathcal{C}} = (0, 1, 0)^{\top}, \\ [T(E_{21})]_{\mathcal{C}} &= [x]_{\mathcal{C}} = (0, 1, 0)^{\top}, \\ [T(E_{22})]_{\mathcal{C}} &= [1 - x^2]_{\mathcal{C}} = (1, 0, -1)^{\top}. \end{aligned}$$

Placing these as columns gives the matrix

$${}_c[T]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix}.$$

Now, suppose we want to analyze this map using bases that reflect the symmetric and anti-symmetric properties of matrices. We define a new basis $\mathcal{B}'$ for V :

$$\mathcal{B}' = (S_1, S_2, S_3, A_1)$$

where $S_1 = E_{11}$, $S_2 = E_{22}$, $S_3 = E_{12} + E_{21}$ (symmetric matrices), and $A_1 = E_{12} - E_{21}$ (anti-symmetric matrix).

We also define a new basis $\mathcal{C}'$ for W :

$$\mathcal{C}' = (1 + x^2, 1 - x^2, x)$$

Evaluating T on the new basis $\mathcal{B}'$ gives

$$T(S_1) = 1 + x^2,$$

$$T(S_2) = 1 - x^2,$$

$$T(S_3) = 2x,$$

$$T(A_1) = 0.$$

These are much nicer outputs; their coordinate vectors in $\mathcal{C}'$ are $(1, 0, 0)^\top$, $(0, 1, 0)^\top$, $(0, 0, 2)^\top$, and $(0, 0, 0)^\top$. The transformation matrix in these new bases is therefore nicer:

$${}_{\mathcal{C}'}[T]_{\mathcal{B}'} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \end{pmatrix}$$

To link these two perspectives mathematically, we construct the change-of-basis matrices by expressing the new bases in terms of the old ones:

$${}_{\mathcal{B}}[I_V]_{\mathcal{B}'} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad {}_{\mathcal{C}'}[I_W]_{\mathcal{C}'} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}$$

By our change-of-basis corollary, the relationship between all these matrices is given by the equation:

$${}_{\mathcal{C}'}[T]_{\mathcal{B}'} = ({}_{\mathcal{C}'}[I_W]_{\mathcal{C}'})^{-1} \cdot {}_{\mathcal{C}'}[T]_{\mathcal{B}} \cdot {}_{\mathcal{B}}[I_V]_{\mathcal{B}'}$$

This demonstrates the power of coordinate isomorphisms: we can rigorously manipulate arbitrary polynomials and 2×2 matrices using nothing but standard 3×4 and 4×4 scalar matrix multiplications.

Corollary 5.19. *Let V be a vector space. For any basis $\mathcal{B}$ of V , the matrix of the identity in that basis is the identity matrix:*

$$_{\mathcal{B}}[I]_{\mathcal{B}} = I.$$

Moreover, if we set

$$P = _{\mathcal{B}}[I]_{\mathcal{B}'},$$

then P is invertible and

$$P^{-1} = _{\mathcal{B}'}[I]_{\mathcal{B}}.$$

That is, the inverse change-of-basis matrix corresponds to swapping $\mathcal{B}'$ and $\mathcal{B}$.

Proof. If $\mathcal{B} = (v_1, \dots, v_n)$, then $I(v_i) = v_i$ and so $[I(v_i)]_{\mathcal{B}} = [v_i]_{\mathcal{B}} = e_i$, so

$$_{\mathcal{B}}[I]_{\mathcal{B}} = I = I_n.$$

Now, for $\mathcal{B}, \mathcal{B}'$ we have, by composition of linear transformations and products of matrices earlier,

$$_{\mathcal{B}}[I]_{\mathcal{B}'} _{\mathcal{B}'}[I]_{\mathcal{B}} = _{\mathcal{B}}[I \circ I]_{\mathcal{B}} = _{\mathcal{B}}[I]_{\mathcal{B}} = I,$$

and the same can be done for $_{\mathcal{B}'}[I]_{\mathcal{B}'} = I$, so by definition $_{\mathcal{B}'}[I]_{\mathcal{B}}$ is invertible and its inverse is $_{\mathcal{B}}[I]_{\mathcal{B}'}$. $\square$

Corollary 5.20. *Let P be the change-of-basis matrix writing source basis $\mathcal{B}'$ in terms of the target basis $\mathcal{B}$, i.e. $P = _{\mathcal{B}}[I]_{\mathcal{B}'}$, and let Q be the change-of-basis matrix writing source basis $\mathcal{C}'$ in terms of the target basis $\mathcal{C}$, i.e. $Q = _{\mathcal{C}}[I]_{\mathcal{C}'}$. Then*

$$_{\mathcal{C}'}[T]_{\mathcal{B}'} = Q^{-1} \cdot _{\mathcal{C}}[T]_{\mathcal{B}} \cdot P.$$

5.3 Image, Kernel and Rank

We now move onto more general properties of linear transformations.

Definition 5.21 (Image and Kernel). Let $T : V \rightarrow W$ be a linear map between vector spaces.

- The *image* (or range) of T is the set of all outputs:

$$\text{im}(T) := T(V) = \{T(v) : v \in V\} \subseteq W.$$

- The *kernel* (or nullspace) of T is the set of all inputs that map to zero:

$$\ker(T) := \{v \in V : T(v) = 0\} \subseteq V.$$

Proposition 5.22. $\operatorname{im}(T)$ and $\ker(T)$ are vector subspaces of W and V , respectively.

Proof. Image: Let $w_1, w_2 \in \operatorname{im}(T)$. By definition, there exist $v_1, v_2 \in V$ such that $T(v_1) = w_1$ and $T(v_2) = w_2$. By linearity:

$$w_1 + w_2 = T(v_1) + T(v_2) = T(v_1 + v_2) \in \operatorname{im}(T)$$

Similarly, for any scalar $\lambda \in F$:

$$\lambda w_1 = \lambda T(v_1) = T(\lambda v_1) \in \operatorname{im}(T).$$

Thus, the image is closed under addition and scalar multiplication.

Kernel: Let $v_1, v_2 \in \ker(T)$. This means $T(v_1) = 0$ and $T(v_2) = 0$. By linearity:

$$T(v_1 + v_2) = T(v_1) + T(v_2) = 0 + 0 = 0 \implies v_1 + v_2 \in \ker(T)$$

For any scalar $\lambda \in F$:

$$T(\lambda v) = \lambda T(v) = \lambda \cdot 0 = 0 \implies \lambda v \in \ker(T)$$

Thus, the kernel is closed under addition and scalar multiplication. $\square$

Observe that $\operatorname{im}(T) = W$ if and only if T is surjective (it has the largest possible image). By linearity, we also have a characterisation for injectivity.

Lemma 5.23. A linear map $T : V \rightarrow W$ is injective if and only if $\ker(T) = \{0\}$.

Proof. Suppose $\ker(T) \neq \{0\}$. Then there exists a non-zero vector $v \neq 0$ such that $T(v) = 0$. Because T is linear, we also know $T(0) = 0$. Since $T(v) = T(0)$ but $v \neq 0$, the map is not injective.

Conversely, suppose T is not injective. There exist distinct vectors $v, w \in V$ ($v \neq w$) such that $T(v) = T(w)$. By linearity:

$$T(v) - T(w) = 0 \implies T(v - w) = 0$$

Since $v \neq w$, the vector $v - w \neq 0$. Thus, $v - w$ is a non-zero vector in the kernel, meaning $\ker(T) \neq \{0\}$. $\square$

Remark. This highlights why the zero vector is so special in linear algebra. Because linear maps always send 0 to 0, we can detect whether a map fails injectivity simply by checking if anything other than 0 gets mapped to 0.

Definition 5.24 (Rank). Let $T : V \rightarrow W$ be a linear map such that $\text{im}(T)$ is finite-dimensional. The *rank* of T is defined as the dimension of its image:

$$\text{rank}(T) := \dim(\text{im}(T)).$$

Example 5.25. • If $T = \lambda I_V : V \rightarrow V$ and V is finite-dimensional, then $\text{rank}(T) = \dim(V)$ unless $\lambda = 0$, in which case $\text{rank}(T) = 0$ since $\text{im}(T) = \{0\}$.

- If $T : V \rightarrow W$ is surjective and W is finite-dimensional, then $\text{rank}(T) = \dim(W)$ because $\text{im}(T) = W$.
- Let $T = \frac{d}{dx} : F[x]_{\leq m} \rightarrow F[x]_{\leq m-1}$ be the formal derivative. Taking the formal derivative of degree m polynomials hits every possible degree $m-1$ polynomial, so the map is surjective. Thus, $\text{rank}(T) = \dim(F[x]_{\leq m-1}) = m$.

However, the map is not injective because the formal derivative sends constants to zero.

- Define $T : F^2 \rightarrow F^2$ by left-multiplication:

$$T \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix}.$$

The image consists of all vectors along the x -axis. This is a 1-dimensional space, so $\text{rank}(T) = 1$.

Remark. The formal derivative example shows that $\frac{d}{dx} : F[x] \rightarrow F[x]$ on the infinite-dimensional space of all polynomials is surjective but not injective (since all constants map to zero). As we will see via the Rank-Nullity Theorem, this behavior (being surjective but not injective) is impossible for a linear map from a finite-dimensional space to itself.

Lemma 5.26. If $T : V \rightarrow W$ is a linear map and $U \subseteq V$ is a subspace such that the restriction $T|_U$ is injective, then $\dim(U) = \dim(T(U))$.

Proof. Restrict the domain to U and the codomain to the image $T(U)$. The restricted map $T|_U : U \rightarrow T(U)$ is surjective by definition. Because we assumed it is also injective, it is a bijection, and therefore a linear isomorphism. By Corollary 5.8, $\dim(U) = \dim(T(U))$. $\square$

Proposition 5.27. 1. Let $A \in M_{m \times n}(F)$ and let $T_A : F^n \rightarrow F^m$ be the left-multiplication map $T_A(v) = Av$. Then $\text{rk}(T_A) = \text{rk}(A)$.
 2. Let $T : V \rightarrow W$ be a linear map between finite-dimensional spaces with bases $\mathcal{B}$ and $\mathcal{C}$. Then $\text{rk}({}_\mathcal{C}[T]_\mathcal{B}) = \text{rk}(T)$.

Proof. (1) The image of T_A is the set of all outputs $\{Av : v \in F^n\}$. By the column-wise definition of matrix multiplication, Av is simply a linear combination of the columns of A :

$$\text{im}(T_A) = \{Av : v \in F^n\} = \text{Span}(Ae_1, \dots, Ae_n)$$

where Ae_i is the i^{th} column of A . The span of the columns is precisely the column space, $\text{CS}(A)$. Since the rank of a matrix is defined as the dimension of its column space:

$$\text{rk}(T_A) := \dim(\text{im}(T_A)) = \dim(\text{CS}(A)) = \text{rk}(A)$$

(2) By definition, the columns of the matrix ${}_ \mathcal{C}[T]_\mathcal{B}$ are the coordinate vectors $[T(v_1)]_\mathcal{C}, \dots, [T(v_n)]_\mathcal{C}$, where the v_i are the basis vectors of $\mathcal{B}$.

The column space of this matrix is the span of those coordinate vectors:

$$\text{CS}({}_\mathcal{C}[T]_\mathcal{B}) = \text{Span}([T(v_1)]_\mathcal{C}, \dots, [T(v_n)]_\mathcal{C}).$$

Because the coordinate mapping $[\cdot]_\mathcal{C}$ is a linear isomorphism, taking the span of the coordinates is the same as taking the coordinates of the span. Thus, this is exactly the coordinate representation of the image space $\text{im}(T)$:

$$\text{CS}({}_\mathcal{C}[T]_\mathcal{B}) = [\text{im}(T)]_\mathcal{C}.$$

Because the coordinate map $[\cdot]_\mathcal{C} : W \rightarrow F^m$ is an isomorphism, it preserves dimension. Therefore, by Lemma 5.26, we have:

$$\dim(\text{CS}({}_\mathcal{C}[T]_\mathcal{B})) = \dim([\text{im}(T)]_\mathcal{C}) = \dim(\text{im}(T)),$$

so $\text{rk}({}_\mathcal{C}[T]_\mathcal{B}) = \text{rk}(T)$, as desired. $\square$

5.4 The Rank-Nullity Theorem

Theorem 5.28 (The Rank-Nullity Theorem). Let $T : V \rightarrow W$ be linear, where V is finite-dimensional. Then:

$$\dim V = \text{rank}(T) + \dim \ker(T).$$

Remark. The core idea relies on extending a basis of a smaller subspace (the kernel). If we simply started with an arbitrary basis of V , say $(v_1, \dots, v_n)$, the list of mapped vectors $(T(v_1), \dots, T(v_n))$ would guarantee a spanning set for $\text{im}(T)$. However, it does not guarantee linear independence because T may not be injective (it might crush some directions together). By ignoring the kernel, we hit a dead end. Instead, we must start by isolating the kernel and build outwards.

Proof. We choose a basis for the kernel

$$(v_1, \dots, v_m),$$

and extend this to a basis of V :

$$(u_1, \dots, u_k, v_1, \dots, v_m).$$

Thus, $\dim V = k + m$. We claim that the list $(T(u_1), \dots, T(u_k))$ is a basis for $\text{im}(T)$.

Linear independence: Suppose a linear combination equals zero:

$$\lambda_1 T(u_1) + \dots + \lambda_k T(u_k) = 0.$$

By linearity, this implies that

$$T(\lambda_1 u_1 + \dots + \lambda_k u_k) = 0,$$

which means the vector $x = \lambda_1 u_1 + \dots + \lambda_k u_k$ lies in $\ker(T)$. But the kernel is spanned entirely by $(v_1, \dots, v_m)$. Because our chosen basis for V is linearly independent, no non-trivial linear combination of the u_i vectors can equal a linear combination of the v_j vectors. Therefore, the only solution is $x = 0$, which forces $\lambda_1 = \dots = \lambda_k = 0$. The set is linearly independent.

Spanning: For any arbitrary $v \in V$, we can express it in our chosen basis:

$$v = \sum_{i=1}^k \alpha_i u_i + \sum_{j=1}^m \beta_j v_j.$$

Applying T and using linearity:

$$T(v) = \sum_{i=1}^k \alpha_i T(u_i) + \sum_{j=1}^m \beta_j T(v_j).$$

Since $v_j \in \ker(T)$, we know $T(v_j) = 0$ for all j . The second sum vanishes completely:

$$T(v) = \sum_{i=1}^k \alpha_i T(u_i),$$

so, every element of $\text{im}(T)$ is a linear combination of $T(u_1), \dots, T(u_k)$, meaning they span $\text{im}(T)$.

Because they are independent and span the space, we have

$$\text{rank}(T) = \dim \text{im}(T) = k.$$

Therefore, $\dim V = k + m = \text{rank}(T) + \dim \ker(T)$. $\square$

5.4.1 The Canonical Block Matrix

The structure of the proof above reveals exactly how we can systematically choose bases to simplify the matrix representation of any linear map.

Let $T : V \rightarrow W$ be linear, let $\mathcal{B} = (u_1, \dots, u_k, v_1, \dots, v_m)$ be our basis for V , where the v_j span the kernel and the u_i fill out the remaining dimensions.

If W is also finite-dimensional, we can take our image basis $(T(u_1), \dots, T(u_k))$ and extend it to a full basis of W :

$$\mathcal{C} = (T(u_1), \dots, T(u_k), w_{k+1}, \dots, w_p)$$

With respect to these specific bases, consider where T sends each basis vector of $\mathcal{B}$:

- $T(u_i)$ is exactly the i^{th} basis vector of $\mathcal{C}$, so the coordinate column has a 1 in the i^{th} position and 0 elsewhere.
- $T(v_j) = 0$, so the coordinate column is entirely zero.

The resulting matrix of T is simplified:

$${}_c[T]_{\mathcal{B}} = \begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix},$$

where I_k is the $k \times k$ identity matrix, and $k = \text{rank}(T)$. There is also the row and column operations viewpoint. Using Gaussian elimination, we can perform row operations to obtain a row echelon form, and then column operations to reach the block matrix

$$\begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix}.$$

So, for any linear map $T : V \rightarrow W$ of rank k , we can always choose suitable bases of V and W such that the linear map can be written as the block matrix

$$\begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix},$$

and conversely, the rank of such a matrix is k , so the rank is the unique integer k such that ${}_c[T]_{\mathcal{B}}$ can be brought to this form, running over all bases $\mathcal{B}$ of V and $\mathcal{C}$ of W .

Remark. This canonical form removes all of the coordinate complexity (i.e. the “messy” matrices we get by choosing random bases) and reveals the “skeleton” of a linear map. It proves that *every* linear transformation, no matter how convoluted its standard matrix entries appear, is fundamentally doing exactly two things:

1. It preserves k dimensions (the I_k block), acting as an isomorphism from the u_i subspace onto the image.
2. It annihilates the remaining dimensions (the zero blocks, representing the kernel).

This provides a complete classification of linear maps. Rank is the *only* true structural invariant. If two linear maps (or matrices) have the same rank, they are geometrically identical transformations—they preserve the exact same amount of information and destroy the exact same amount of information. Any apparent difference between them is merely an artifact of looking at them through misaligned or skewed coordinate bases.

As a corollary we get:

Corollary 5.29. *Let $A \in M_{m,n}(F)$ and let $T_A : F^n \rightarrow F^m$ be the associated linear map. Then there exist invertible matrices $Q \in GL_m(F)$, $P \in GL_n(F)$ such that*

$$Q^{-1}AP = \begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix},$$

where $k = \text{rank}(A)$. In other words, A is equivalent to this canonical form under row and column operations.

Proof. This follows immediately from the change of basis formula (Theorem 5.16) and the above discussion. $\square$

In other words, we have

$$\{A \in M_{m \times n}(F) : \text{rank}(A) = k\} = \left\{ Q \begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix} P^{-1} : Q \in GL_m(F), P \in GL_n(F) \right\}.$$

5.4.2 Left and Right Inverses

Another nice consequence of the Rank-Nullity Theorem is the following fact:

Corollary 5.30. *Let $T : V \rightarrow W$ be linear, with V and W finite-dimensional. Then the following are equivalent:*

1. T is bijective (an isomorphism),
2. T is injective and $\dim V = \dim W$,
3. T is surjective and $\dim V = \dim W$.

Proof. (1) $\Rightarrow$ (2), (3): Suppose T is an isomorphism, then in particular it is injective and surjective. By the lemma above, $\dim V = \dim W$.

(2) $\Rightarrow$ (1): If T is injective then $\ker(T) = \{0\}$ so by rank-nullity

$$\dim V = \text{rk}(T) + \dim \ker(T) = \text{rk}(T).$$

But $\dim V = \dim W$ so $\text{rk}(T) = \dim W$. Hence $\text{im}(T)$ is a subspace of W of full dimension, so $\text{im}(T) = W$, i.e. T is surjective as well. Thus T is bijective.

(3) $\Rightarrow$ (1): If T is surjective then $\text{im}(T) = W$, so

$$\text{rk}(T) = \dim W = \dim V.$$

By rank-nullity,

$$\dim V = \text{rk}(T) + \dim \ker(T),$$

so this forces $\dim \ker(T) = 0$, so as we saw in problem sheets this implies $\ker(T) = \{0\}$ and thus T is injective, so T is bijective. $\square$

We can also interpret this in terms of left and right inverses:

Proposition 5.31. *Given a linear transformation $T : V \rightarrow W$ for V and W finite-dimensional:*

1. *There exists a left inverse $S : W \rightarrow V$, i.e. such that $S \circ T = I_V$ if and only if T is injective,*
2. *There exists a right inverse $S : W \rightarrow V$, i.e. such that $T \circ S = I_W$, if and only if T is surjective.*

Proof. 1. First, if there exists a left inverse then $S \circ T = I_V$ is injective, so T must also be injective. Conversely, if T is injective then $\ker(T) = \{0\}$ so pick a basis $B = (v_1, \dots, v_n)$ of V . We then have that $Tv_1, \dots, Tv_n$ is linearly independent and can be extended to a basis $(Tv_1, \dots, Tv_n, w_1, \dots, w_{m-n})$ of W . We then define $S(Tv_i) = v_i$ and $S(w_i) = 0$ and this uniquely extends to a linear map $S : W \rightarrow V$ with the property that $S(Tv_i) = v_i$ for all i and since $(v_1, \dots, v_n)$ is a basis, $S \circ T = I_V$, as desired.

2. First, if there exists a right inverse then $T \circ S = I_W$ is surjective, so T must also be surjective. Conversely, if T is surjective, pick a basis $B = (v_1, \dots, v_n, u_1, \dots, u_k)$ of V such that $(u_1, \dots, u_k)$ is a basis for $\ker(T)$. So $(Tv_1, \dots, Tv_n)$ is a basis of C , as above. We can then define $S(Tv_i) = Tv_i$ for all i and this uniquely extends to a linear map $S : W \rightarrow V$ with the property that $T \circ S(Tv_i) = Tv_i$ for all i , so $T \circ S = I_W$ since $(Tv_1, \dots, Tv_n)$ is a basis of W . $\square$

Corollary 5.32. *Let $T : V \rightarrow W$ be linear, with V and W finite-dimensional. Then the following are equivalent:*

1. T is bijective (i.e. an isomorphism),
2. T has a left inverse and $\dim V = \dim W$,
3. T has a right inverse and $\dim V = \dim W$.

Proof. Follows from Corollary 5.30 and Proposition 5.31. $\square$

Remark. As before, if T is invertible, any left inverse is the inverse, and any right inverse is the inverse. Taking matrices, we recover what we knew before, that square matrices have an inverse if and only if they have *either* a left or a right inverse (and, as we saw, every left or right inverse in this case is an inverse).

Chapter 6

Dual Vector Spaces

Definition 6.1. Let V be a vector space over F . The dual space, denoted V^* , is the vector space of all linear maps from V to F :

$$V^* := \{T : V \rightarrow F \mid T \text{ is linear}\}$$

Remark. A vector $v \in V$ is an algebraic object (a geometric arrow, a polynomial, a matrix). A dual vector $f \in V^*$ is a linear scalar-valued function. It is best understood as an abstract “measurement object” that takes a vector and returns a scalar value in the base field F , representing some linear property of that vector.

If $V = F^n$ consists of column vectors, V^* consists of row vectors. The evaluation $f(v)$ is exactly the matrix multiplication of a $1 \times n$ row vector with an $n \times 1$ column vector to yield a 1×1 scalar, i.e. the dot product (we will see what this means geometrically in $\mathbb{R}^n$ later).

Example 6.2. Let $V = \mathbb{R}[x]_{\leq 2}$ be the vector space of polynomials in $\mathbb{R}$ of degree at most 2. The elements are polynomials $p(x) = a + bx + cx^2$. We can define several distinct dual vectors (“measurement objects”) in V^* :

- Pointwise evaluation: Let $E_2 \in V^*$ be defined by $E_2(p) = p(2)$. This function takes a polynomial and returns its value at $x = 2$. It is linear, so $E_2 \in V^*$.
- Differentiation: Let $D_0 \in V^*$ be defined by $D_0(p) = p'(0)$. This measures the slope of the polynomial at the origin (which evaluates to the coefficient b).
- Integration: Let $I \in V^*$ be defined by $I(p) = \int_0^1 p(x)dx$. This measures the area under the curve.

Each of these is a distinct vector in the dual space V^* .

The vector space structure on V^* is given by pointwise addition and scalar multiplication. For $f, g \in V^*$ and $\lambda \in F$:

$$\begin{aligned}(f + g)(v) &= f(v) + g(v) \\ (\lambda f)(v) &= \lambda(f(v))\end{aligned}$$

This space is also denoted as $\text{Hom}(V, F)$. More generally, $\text{Hom}(V, W)$ denotes the vector space of linear maps from V to W .

Exercise. Verify the vector space axioms for $\text{Hom}(V, W) := \{T : V \rightarrow W \mid T \text{ is linear}\}$, and hence also for V^* .

Example 6.3. Consider $V = F^n$, viewed as column vectors. All linear maps $V \rightarrow F \cong F^1$ are given by multiplication by matrices, in this case row vectors $u \in M_{1 \times n}(F)$. We can identify the dual with row vectors $M_{1 \times n}(F)$, so that if $u \in M_{1 \times n}(F)$ and $v \in M_{n \times 1}(F)$, we get the linear transformation $T_u : M_{n \times 1}(F) \rightarrow M_{1 \times 1}(F) \cong F$. Precisely, this gives an isomorphism $M_{n \times 1}(F) \rightarrow M_{1 \times n}(F)^*$, $u \mapsto T_u$.

The preceding example shows that dualising a finite-dimensional vector space produces another vector space of the same dimension ($\dim V = n \implies V \cong F^n \implies V^* \cong (F^n)^*$).

6.1 Dual Transformations

In addition to dualising vector spaces, we can also dualise maps.

Definition 6.4. Given a linear transformation $T : V \rightarrow W$, the dual transformation $T^* : W^* \rightarrow V^*$ is defined as:

$$T^*(f) := f \circ T$$

Remark. Given a linear map $T : V \rightarrow W$, the dual map $T^* : W^* \rightarrow V^*$ reverses the direction. When translated into matrices via bases, this exact operation yields the transpose: ${}_{\mathcal{B}^v}[T^*]_{\mathcal{C}^w} = ({}_{\mathcal{C}}[T]_{\mathcal{B}})^{\top}$, which is the “dual” idea (we will see why this is the case later).

More specifically, if you want to evaluate an object $v \in V$ using a measuring object $f \in W^*$ after a transformation T has been applied, there are two “dual” perspectives. You can either push $v \in V$ forward via T , and then measure it with f . This is $f(T(v))$. Alternatively, the “dual” perspective is that you can pull f backwards via T^* , and measure the original $v \in V$ using a measurement object in V^* , namely $T^*(f)$. Formally, we say that T^* is the pullback induced by T .

Proposition 6.5. *Given a linear transformation T , the dual T^* is also a linear transformation.*

Proof. We have $T^*(f + g) = (f + g) \circ T = (f \circ T) + (g \circ T) = T^*f + T^*g$, and also $T^*(\lambda f) = (\lambda f) \circ T = \lambda(f \circ T) = \lambda T^*(f)$. $\square$

Example 6.6. Let $T = T_A : F^n \rightarrow F^m$. Identifying $F^n \cong M_{n \times 1}(F)$ and $(F^n)^* \cong M_{1 \times n}(F)$, we have $T_A^*(u)v = u(Av)$, which implies $T_A^*u = uA$. Thus, $T_A^* : M_{1 \times m}(F) \rightarrow M_{1 \times n}(F)$ is the operation of right multiplication by A on row vectors.

Proposition 6.7. *Given a composition $U \xrightarrow{S} V \xrightarrow{T} W$, we have the corresponding dual maps $U^* \xleftarrow{S^*} V^* \xleftarrow{T^*} W^*$, and:*

$$(T \circ S)^* = S^* \circ T^*$$

Also, $I_V^ = I_{V^*}$.*

Proof. The first assertion is associativity of composition: $(T \circ S)^*(f) = f \circ (T \circ S) = (f \circ T) \circ S = S^*(T^*(f))$. The second follows from the identity for composition: $f \circ I_V = f$. $\square$

Corollary 6.8. *If $T : V \rightarrow W$ is invertible, so is T^* , and $(T^{-1})^* = (T^*)^{-1}$.*

Proof. We have $T^* \circ (T^{-1})^* = (T^{-1} \circ T)^* = I_V^* = I_{V^*}$. Similarly $(T^{-1})^* \circ T^* = I_W^* = I_{W^*}$. $\square$

6.2 Dual Basis

Definition 6.9. Given a finite basis $(v_1, \dots, v_n)$ of a vector space V over F , let $v_1^\vee, \dots, v_n^\vee \in V^*$ be the functions defined by:

$$v_i^\vee(\lambda_1 v_1 + \dots + \lambda_n v_n) = \lambda_i$$

In particular, we have the formula:

$$v_i^\vee(v_j) = \delta_{ij} := \begin{cases} 1, & \text{if } i = j \\ 0, & \text{otherwise} \end{cases}$$

Proposition 6.10. *The n -tuple $(v_1^\vee, \dots, v_n^\vee)$ of elements of V^* is a basis.*

Proof. First we show that the tuple is linearly independent. If $f := \lambda_1 v_1^\vee + \dots + \lambda_n v_n^\vee = 0$, then evaluating at v_i gives $f(v_i) = \lambda_i = 0$ for all i .

Next we show that it spans. If $f \in V^*$, we claim that:

$$f = f(v_1)v_1^\vee + \dots + f(v_n)v_n^\vee \quad (6.1)$$

Let g be the right-hand side. Then $g(v_i) = f(v_i)$ for all i , by definition. By linearity and the fact that $(v_1, \dots, v_n)$ span V , we get $g = f$. $\square$

Definition 6.11. The basis $\mathcal{B}^\vee := (v_1^\vee, \dots, v_n^\vee)$ is called the *dual basis* of $\mathcal{B} = (v_1, \dots, v_n)$.

This yields the companion formula for all $v \in V$:

$$v = v_1^\vee(v)v_1 + \dots + v_n^\vee(v)v_n.$$

Corollary 6.12. *For V a finite-dimensional vector space, V^* is also finite-dimensional, and*

$$\dim V^* = \dim V$$

This together with the injectivity of ι yields:

Corollary 6.13. *For V a finite-dimensional vector space, $\iota : V \rightarrow (V^*)^*$ is an isomorphism.*

Proof. We have seen ι is linear and injective. Since V is finite-dimensional, so is $(V^*)^*$, and it has the same dimension. Now ι is a linear map between finite-dimensional vector spaces of the same dimension; so it is an isomorphism. $\square$

6.2.1 The Double Dual

Given a vector space V , we define the map:

$$\iota : V \rightarrow (V^*)^*, \quad v \mapsto \iota_v, \quad \text{where } \iota_v(f) = f(v)$$

Remark. Note that no basis is required in order to define this map.

Proposition 6.14. *The map ι is injective and linear (assuming V is finite-dimensional).*

Proof. Linearity follows because $\iota_{v+v'}(f) = f(v + v') = f(v) + f(v') = \iota_v(f) + \iota_{v'}(f)$, and similarly $\iota_{\lambda v}(f) = \lambda \iota_v(f)$.

For injectivity, suppose $v \in V$ is nonzero. Then we have an inclusion $F \cdot v = \text{Span}(v) \subseteq V$ of subspaces, so the restriction map $V^* \rightarrow (F \cdot v)^*$ is surjective. That is, there exists $f \in V^*$ with $f(v) \neq 0$. Then $\iota_v(f) = f(v) \neq 0$ as well, so $\iota_v \neq 0$. Thus, $\ker(\iota) = \{0\}$, so ι is injective. $\square$

We will see that ι is an isomorphism when V is finite-dimensional. To do this, we find a basis for V^* .

Given a finite-dimensional vector space V and a basis $\mathcal{B} \subseteq V$, we form the bases:

$$\mathcal{B}^\vee \subseteq V^*, \quad \mathcal{B}^{\vee\vee} \subseteq V^{**}$$

We also have the isomorphism $\iota : V \xrightarrow{\sim} (V^*)^*$.

Proposition 6.15. *Taking the double dual basis recovers the image under the isomorphism ι :*

$$\mathcal{B}^{\vee\vee} = \iota(\mathcal{B})$$

Proof. Let $\mathcal{B} = (v_1, \dots, v_n)$, $\mathcal{B}^\vee = (v_1^\vee, \dots, v_n^\vee)$, and $\mathcal{B}^{\vee\vee} = (v_1^{\vee\vee}, \dots, v_n^{\vee\vee})$. Then we have:

$$\iota(v_i)(v_j^\vee) = v_j^\vee(v_i) = \delta_{ij} = v_i^{\vee\vee}(v_j^\vee)$$

Fix i . Since $\iota(v_i)(v_j^\vee) = v_i^{\vee\vee}(v_j^\vee)$ for all j , by taking linear combinations of the basis $(v_1^\vee, \dots, v_n^\vee)$ of V^* , we get $\iota(v_i)(f) = v_i^{\vee\vee}(f)$ for all $f \in V^*$. That is, $\iota(v_i) = v_i^{\vee\vee}$. $\square$

As a consequence, $v_i^{\vee\vee}$ depends only on v_i , not on v_j for $j \neq i$. This is not true about the dual basis: $v_i^\vee$ depends on all v_j . It is defined so that $v_i^\vee(v_j) = 0$ when $i \neq j$.

Example 6.16. Let $V = F^2$, with basis $\mathcal{B} = (v_1, v_2)$ for $v_1 = e_1$ and $v_2 = \lambda e_1 + e_2$. Then $v_1^\vee = e_1^\vee - \lambda e_2^\vee$. We see that it depends on λ , hence on v_2 . In terms of column and row vectors: $\mathcal{B} = \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} \lambda \\ 1 \end{pmatrix} \right)$ and $\mathcal{B}^\vee = \left((1 \ 0), (-\lambda \ 1) \right)$.

6.2.2 Annihilators

Definition 6.17. Given $W \subseteq V$, we define $W^\perp \subseteq V^*$ by:

$$W^\perp := \{f \in V^* \mid f(w) = 0, \forall w \in W\}$$

In other words, it is the set of vectors which annihilate all of W .

Example 6.18. Suppose V has basis $\mathcal{B} = (v_1, \dots, v_n)$ and $W = \text{Span}(v_1, \dots, v_k)$ for some $1 \leq k \leq n$. Then $W^\perp = \text{Span}(v_{k+1}^\vee, \dots, v_n^\vee)$. For example, if $V = M_{n \times 1}(F)$ and $v_i = e_i$ is the standard basis, then

$$W = \left\{ \begin{pmatrix} a_1 \\ \vdots \\ a_m \\ 0 \\ \vdots \\ 0 \end{pmatrix} : a_1, \dots, a_m \in F \right\}, \quad W^\perp = \{(0 \quad \dots \quad 0 \quad b_1 \quad \dots \quad b_{n-m}) : b_1, \dots, b_{n-m} \in F\}$$

Proposition 6.19. If V is finite-dimensional, and $W \subseteq V$ a subspace, then:

$$\dim V = \dim W + \dim W^\perp$$

Remark. When inner products are introduced, they provide an isomorphism $V \rightarrow V^*$, and using this, the preimage of $W^\perp$ is the usual subspace of vectors which dot with all $w \in W$ to zero.

6.3 Dualisation and Transpose

Theorem 6.20. Let V and W be finite-dimensional vector spaces with bases $\mathcal{B}$ and $\mathcal{C}$, respectively. Let $T : V \rightarrow W$ be a linear transformation. Then:

$${}_{\mathcal{B}^\vee}[T^*]_{\mathcal{C}^\vee} = ({}_c[T]_{\mathcal{B}})^\top$$

Proof. Let $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{C} = (w_1, \dots, w_m)$, with dual bases $\mathcal{B}^\vee = (v_1^\vee, \dots, v_n^\vee)$ and $\mathcal{C}^\vee = (w_1^\vee, \dots, w_m^\vee)$. Then

$$T^*(w_j^\vee)(v_i) = w_j^\vee \circ T(v_i) = w_j^\vee(T(v_i))$$

By definition, the left-hand side is the ij -th entry of the matrix ${}_{\mathcal{B}^\vee}[T^*]_{\mathcal{C}^\vee}$, whereas the right-hand side is the ji -th entry of the matrix ${}_c[T]_{\mathcal{B}}$. $\square$

Example 6.21. In the case $V = F^n, W = F^m$, we can verify the theorem directly. Every linear transformation from V to W is of the form T_A for $A \in M_{m \times n}(F)$. We know the formula ${}_{(e_1, \dots, e_m)}[T_A]_{(e_1, \dots, e_n)} = A$. If we identify F^n with column vectors, and $(F^n)^*$ with row vectors, then for $u \in M_{1 \times n}(F)$ a row vector, thought of as an element of V^* , we have

$$[u]_{(e_1^\vee, \dots, e_n^\vee)} = u^\top$$

We found previously that $T_A^*(T_u) = T_{uA}$, left multiplication by the row vector $uA \in M_{1 \times n}(F) \cong V^*$. We then get:

$$[T_A^*u]_{(e_1^\vee, \dots, e_n^\vee)} = [uA]_{(e_1^\vee, \dots, e_n^\vee)} = (uA)^\top = A^\top u^\top = A^\top [u]_{(e_1^\vee, \dots, e_n^\vee)}$$

This implies ${}_{B^\vee}[T_A^*]_{C^\vee} = A^\top$, by the definition of the matrix of a linear transformation.

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